

Speaker diarization

a love story

Hervé Bredin



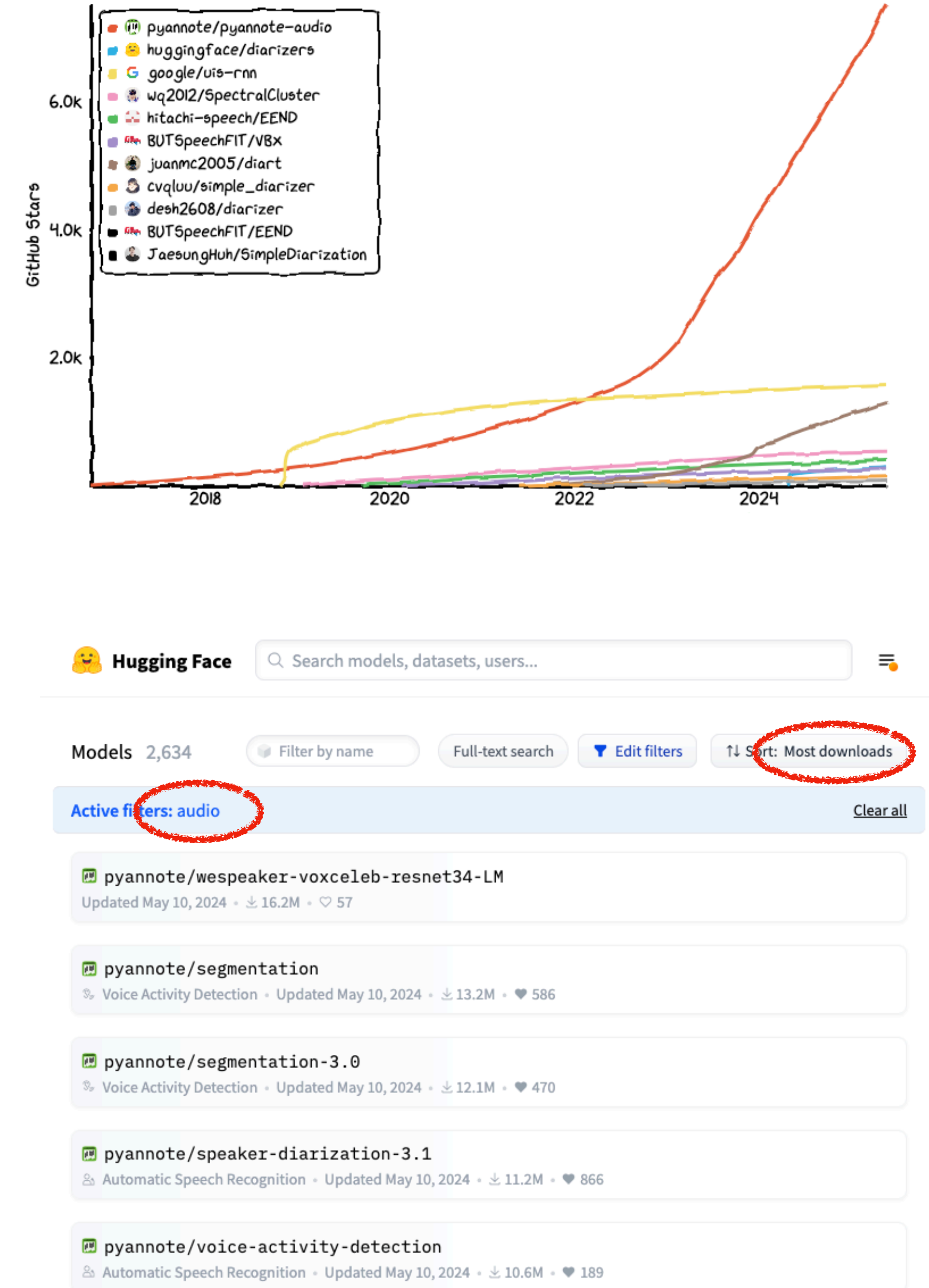
pyannoteAI



Hervé Bredin

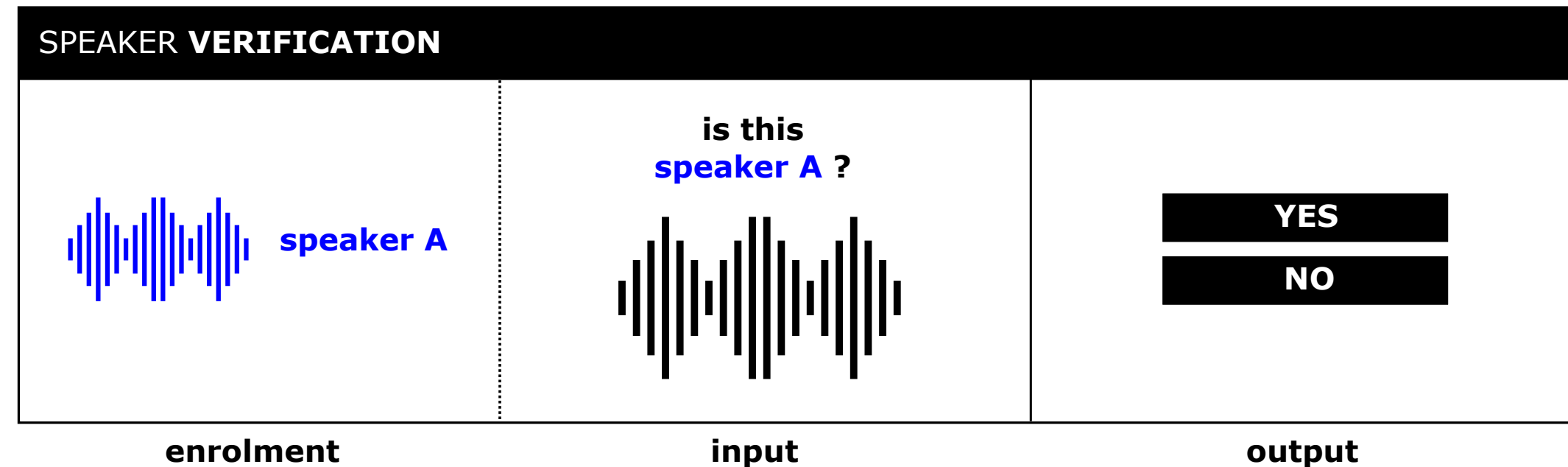
the "pyannote" guy

- Tenured **CNRS** researcher since 2008
- Started working on **speaker diarization** back in 2012 (and never looked back)
- Started open-sourcing **pyannote** toolkit in 2015 (and don't regret it)
- Co-founded **pyannoteAI** in 2024 (on leave from CNRS since 2025)

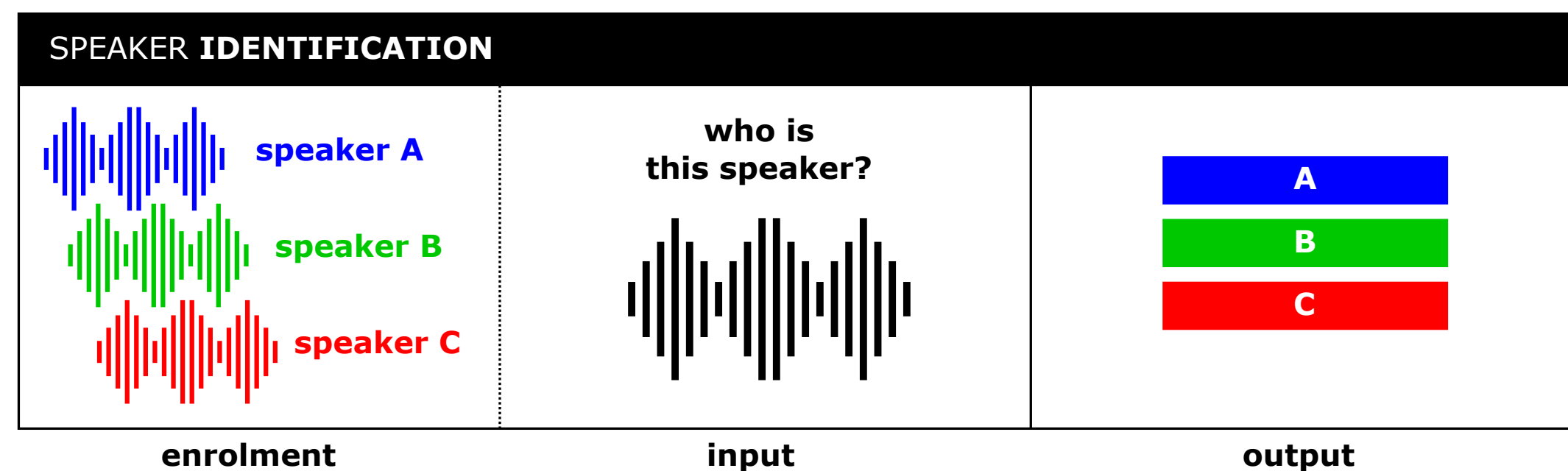


What is speaker diarization?

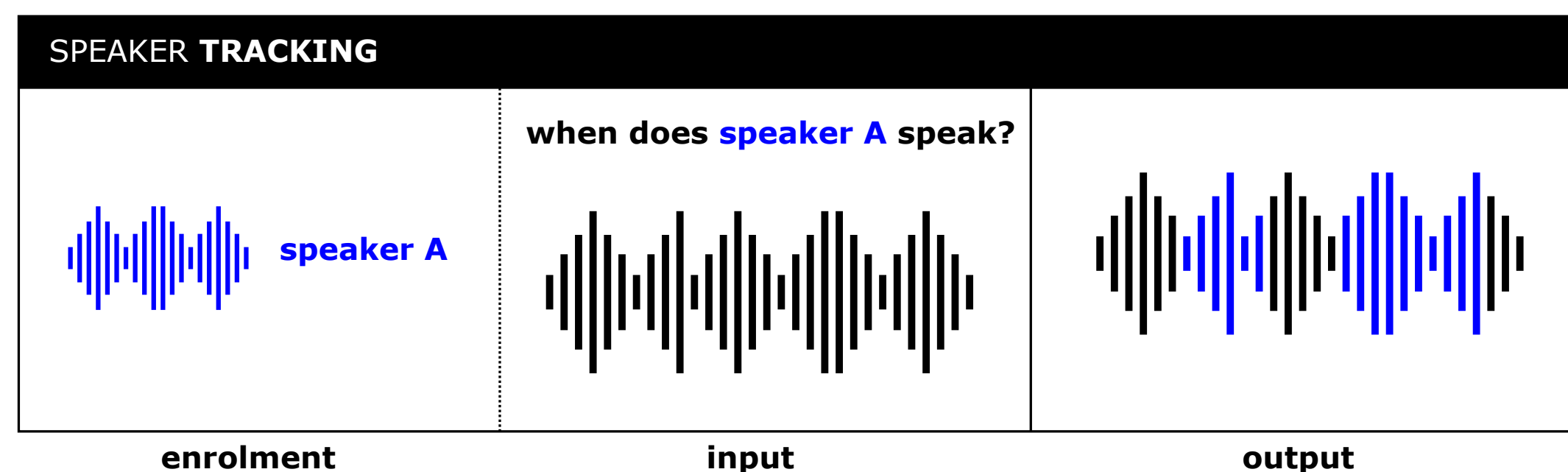
Speaker recognition



*supervised, utterance-wise
binary classification*



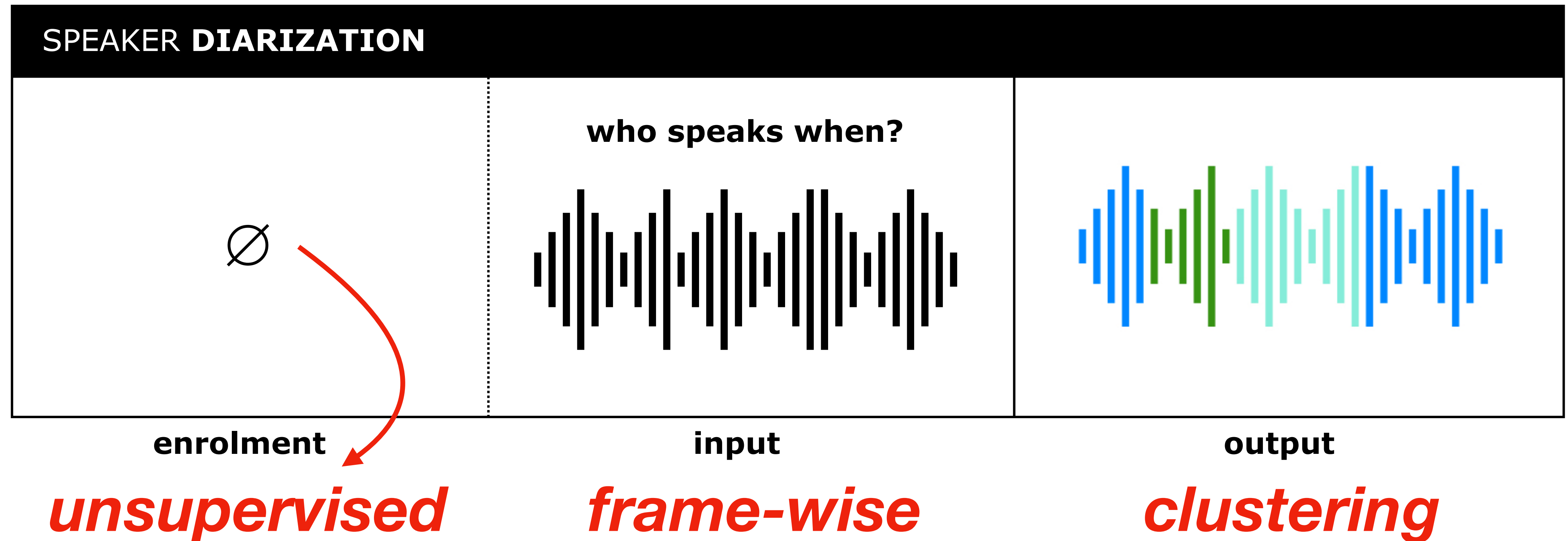
*supervised, utterance-wise
multi-class classification*



*supervised, frame-wise
binary classification*

Speaker diarization

"who speaks when"



Speaker diarization(s)

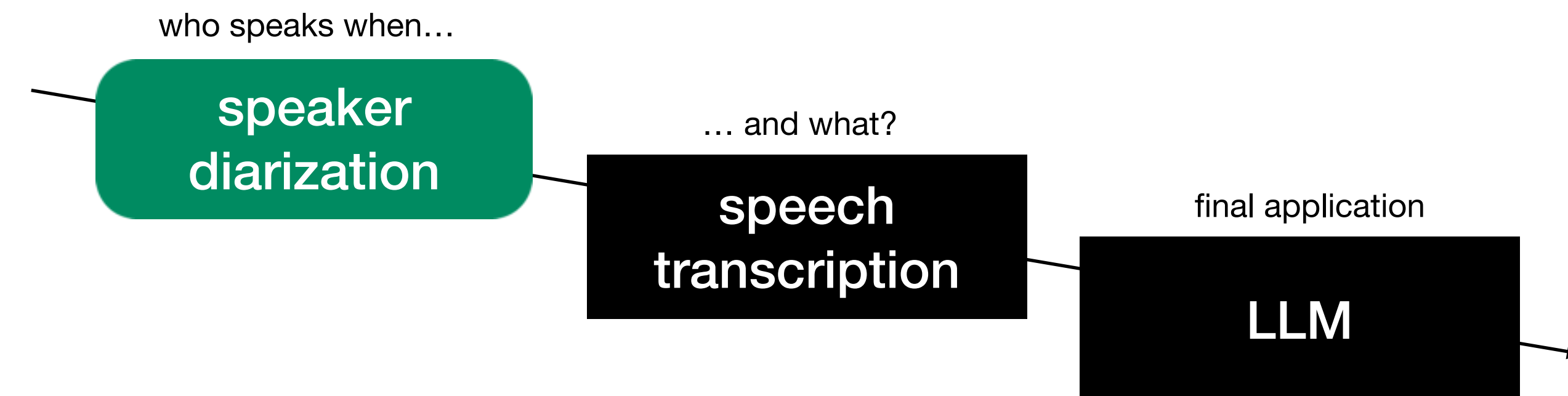
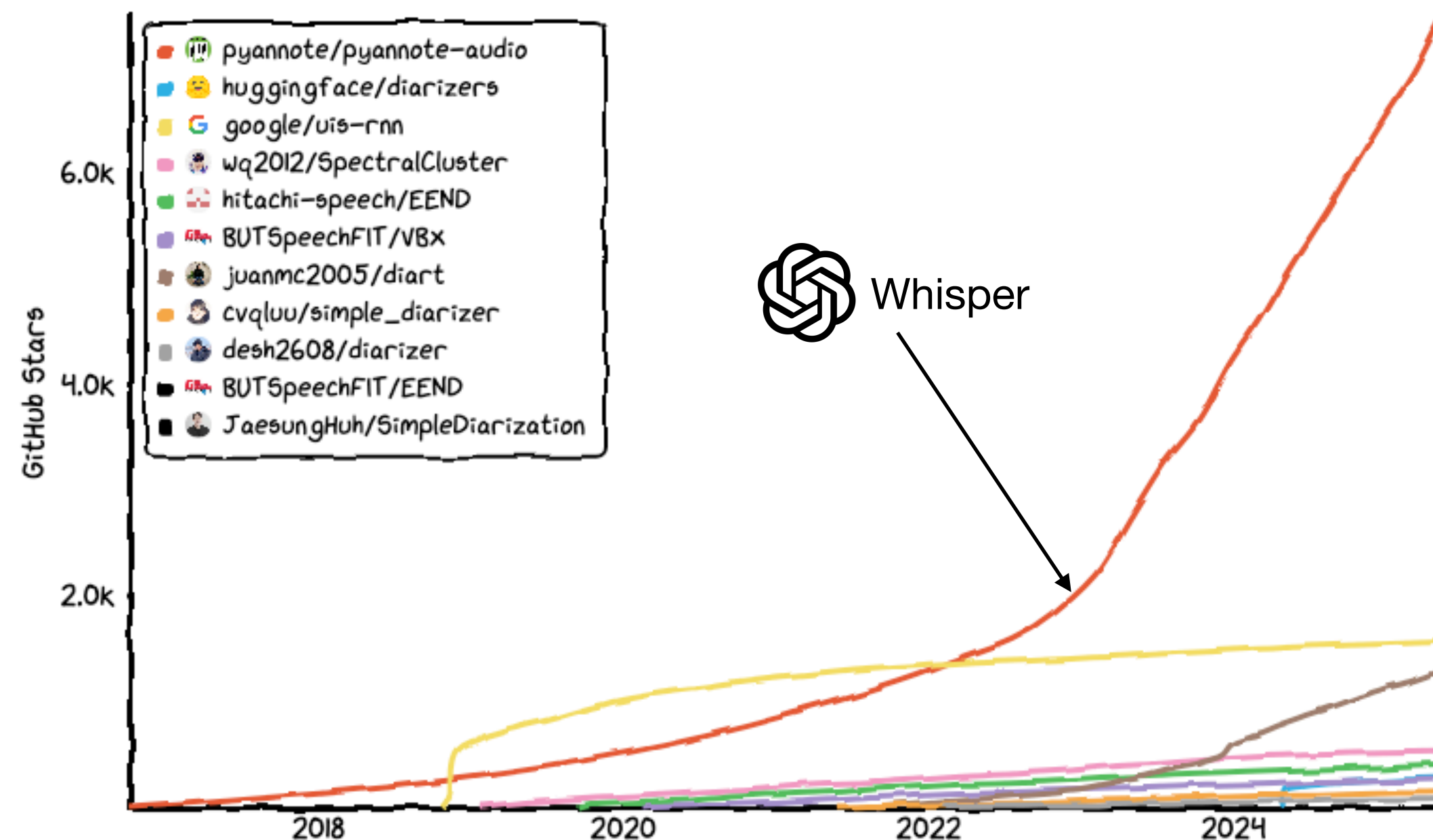
- **Batch** speaker diarization
assumes the whole conversation is available at once
- **Streaming** speaker diarization
processes the conversation in "real-time" as it unfolds
- **Longitudinal** speaker diarization
tracks speakers across multiple conversations

What for?

"a means to an end"

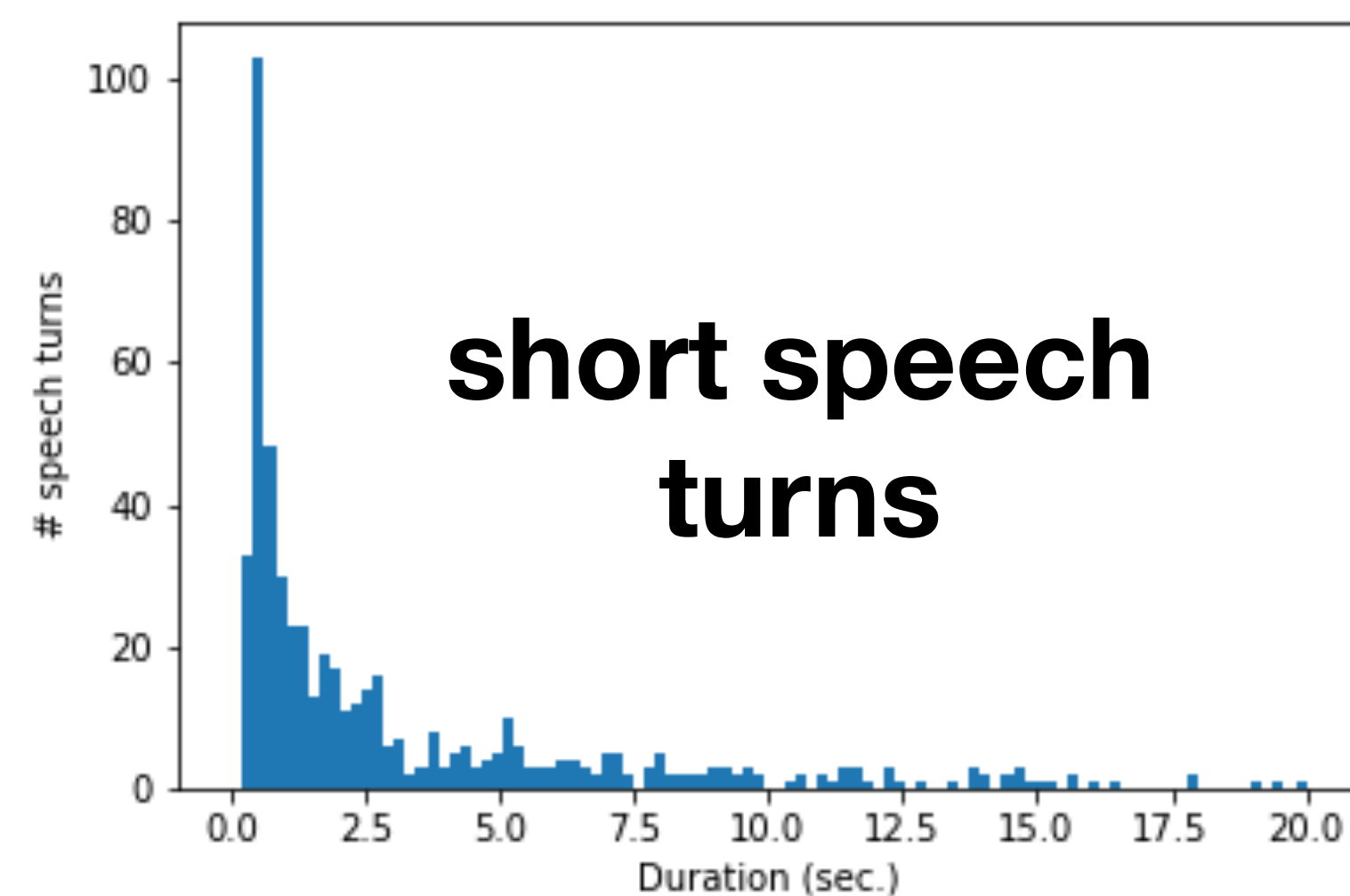
Speaker diarization is an **enabling technology** for **downstream applications**.

- Speaker-attributed transcription
meeting, call center, healthcare, court



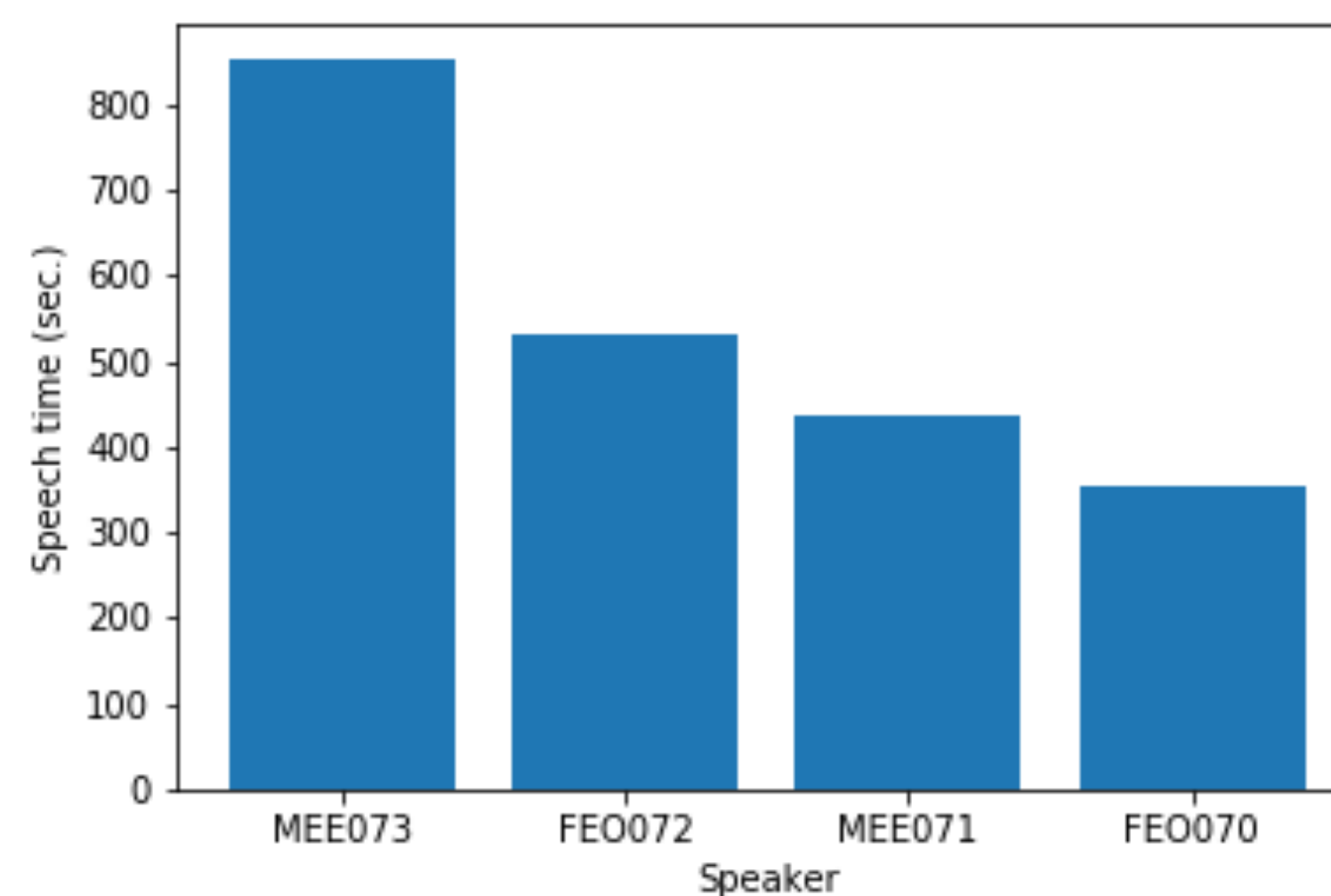
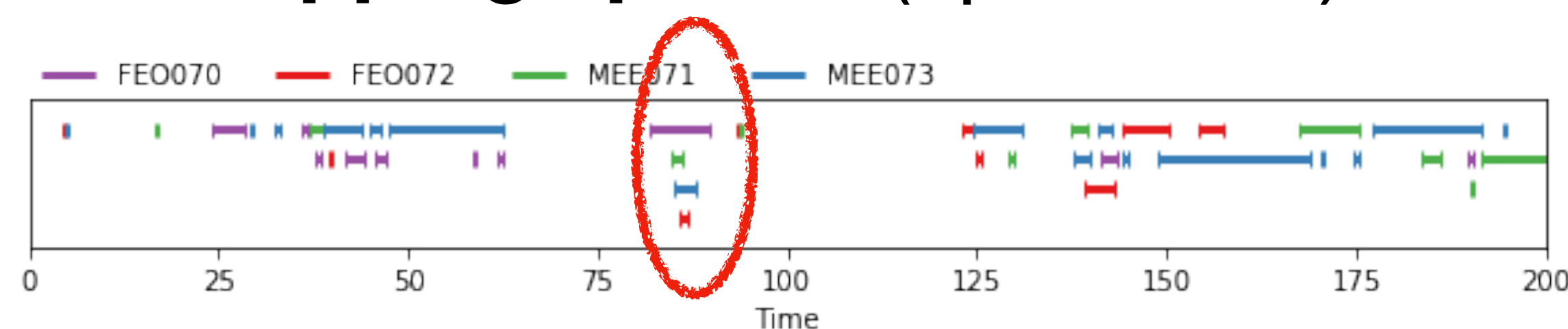
- Voice assistant training  Moshi
- Text-to-speech training  Audiobox
- Video translation with voice cloning
- Life logging (wearables)

What makes it difficult?



**unknown number
of speakers**

overlapping speech (up to 40%)

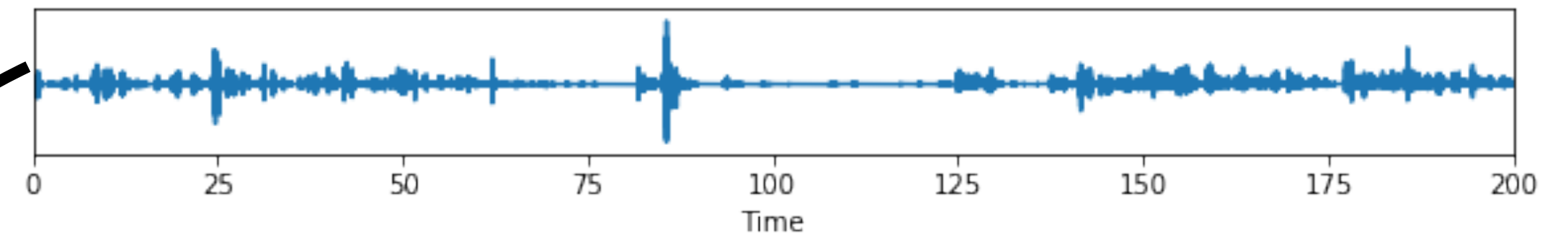


**unbalanced
speaker
time**

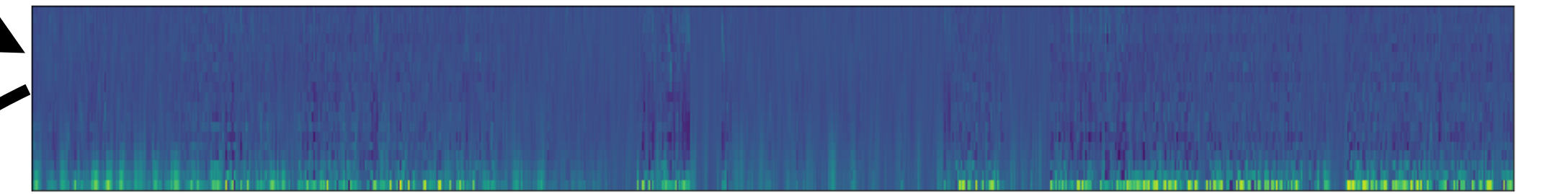
How does it work?

"Historical" approach

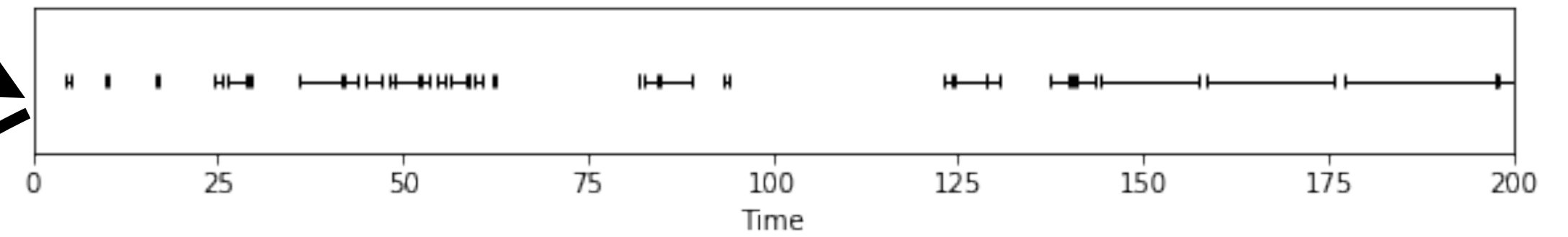
feature extraction



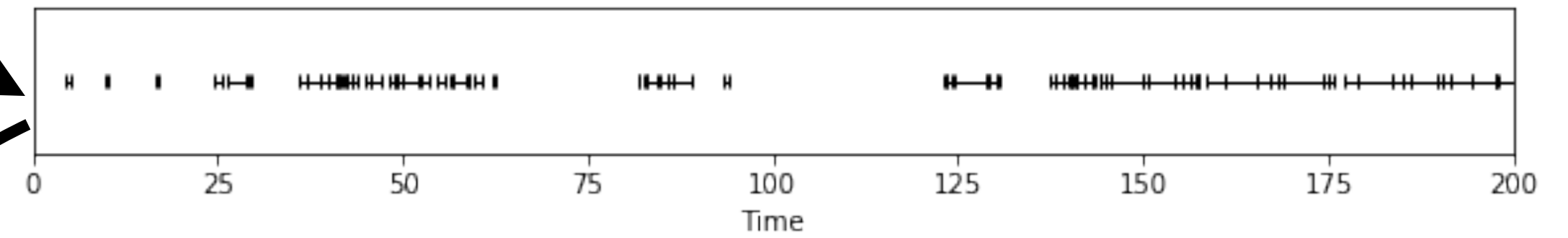
speech activity detection



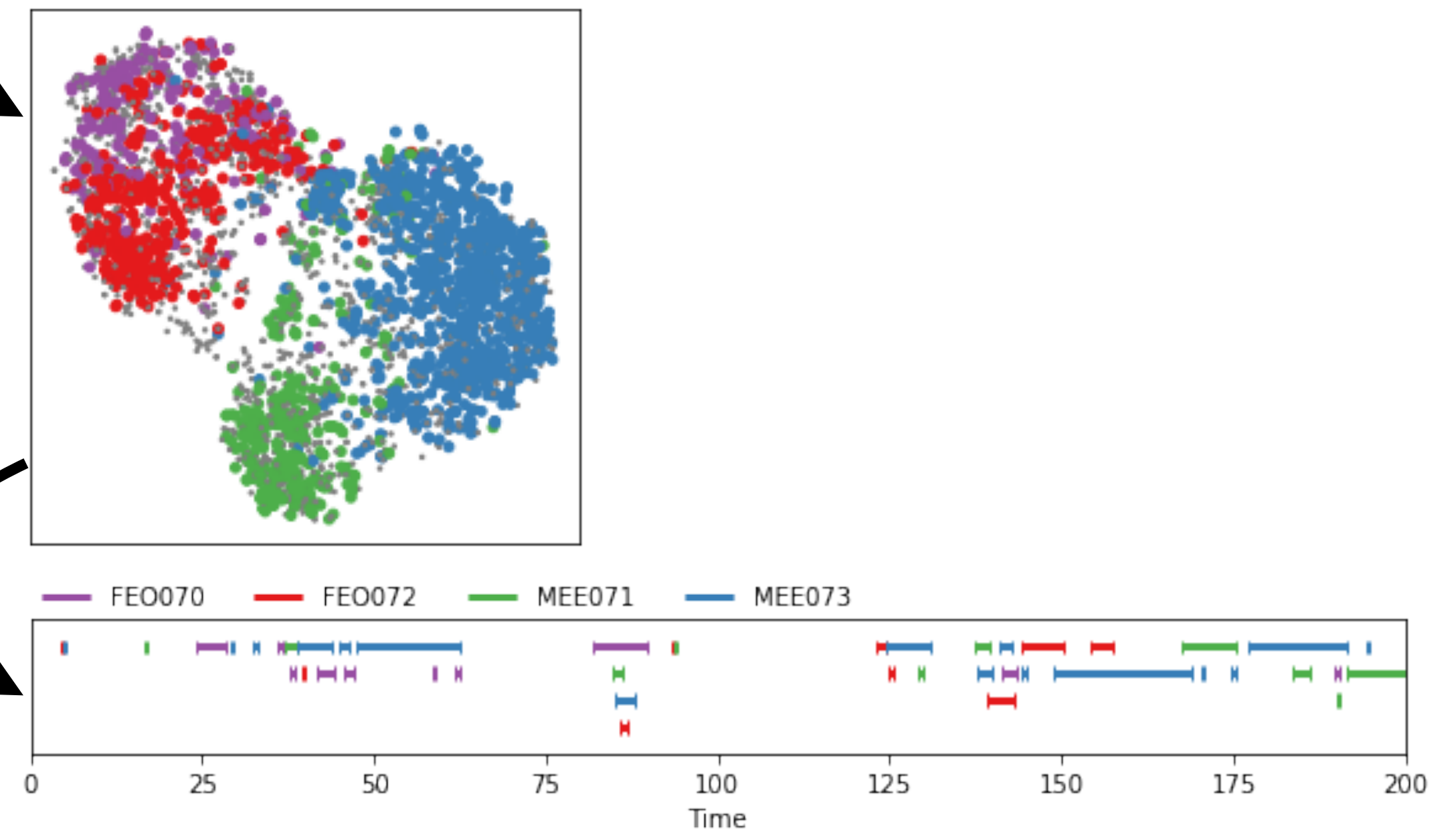
speaker change detection



speech turn representation

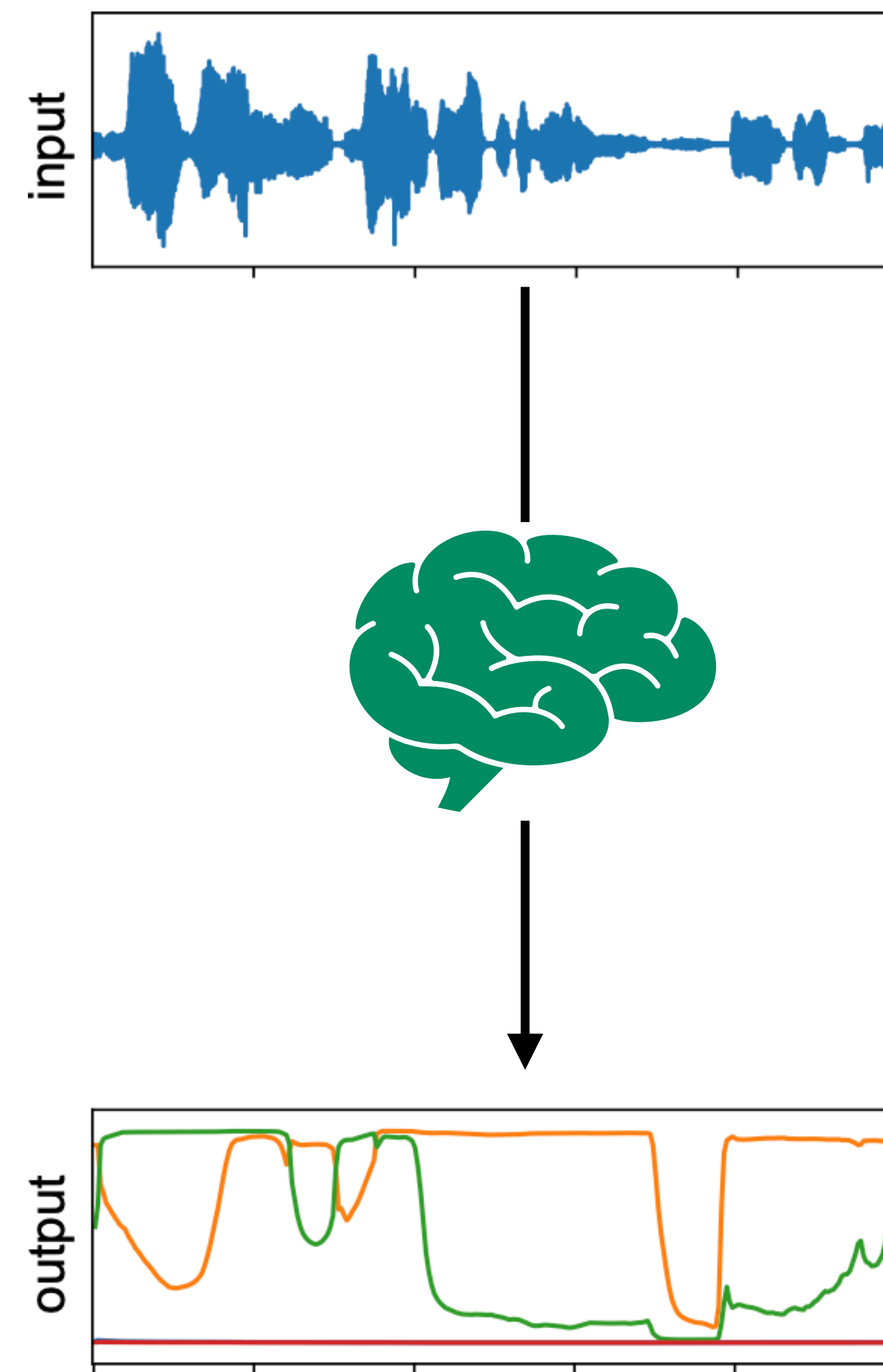


speech turn clustering



End-to-End Neural Diarization

- **Sequence-to-sequence**
input: audio
output: speaker probabilities
- **Multi-label classification**
fixed number of speakers
overlap-aware by design



Pros and cons

Multi-Stage Diarization

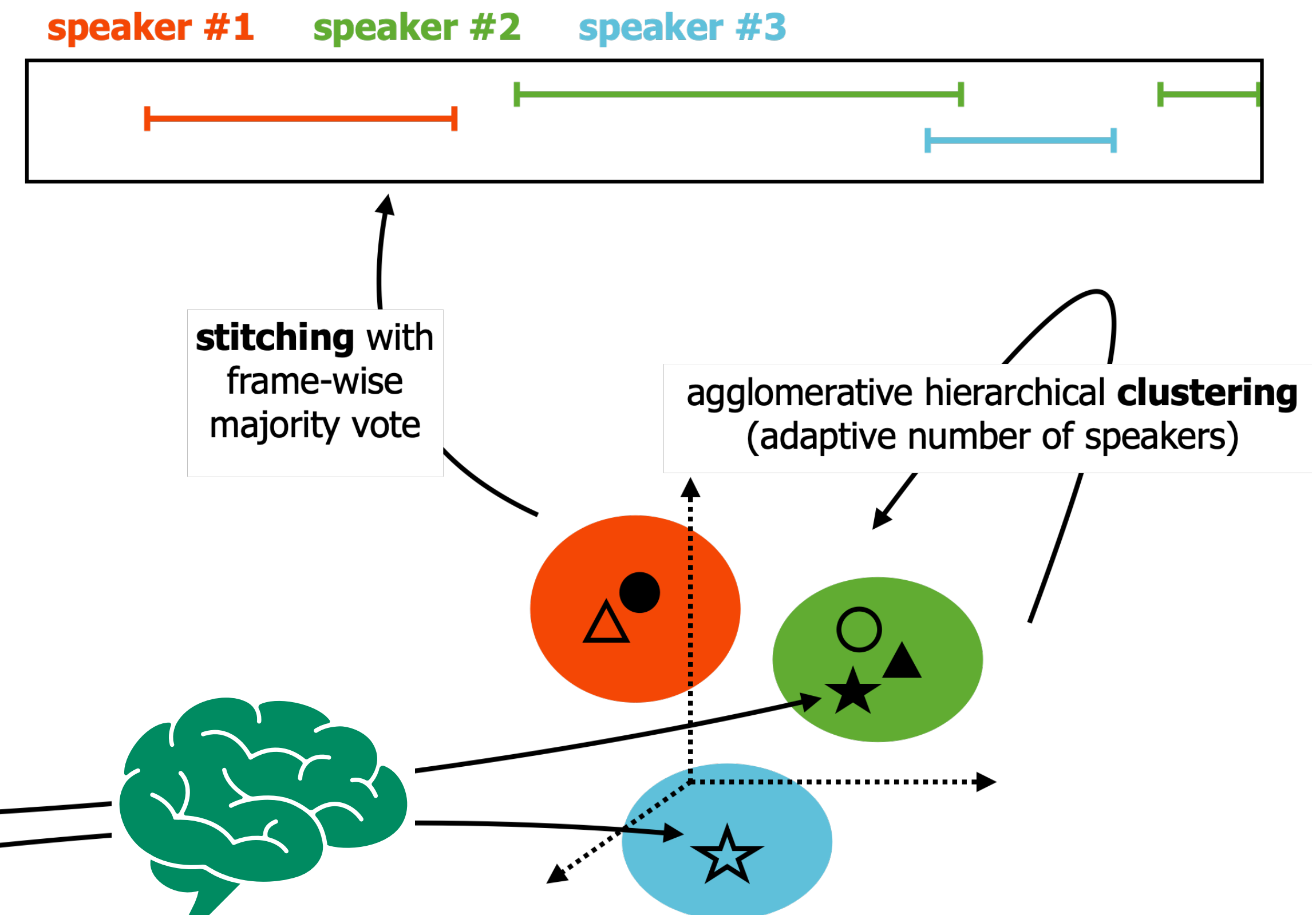
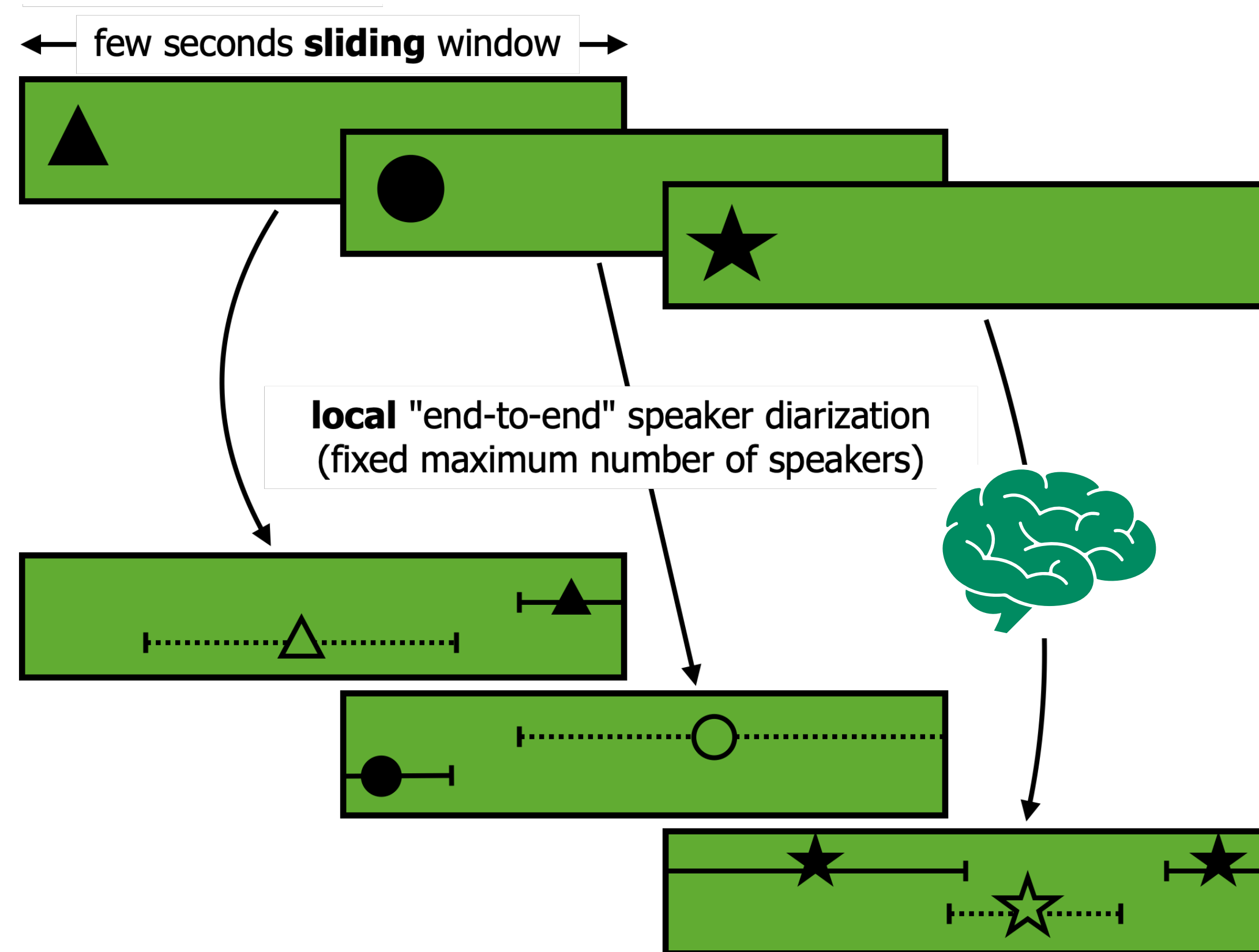
- ✗ propagation of errors
- ✗ hacky handling of overlapping speech
- ✓ (relatively) easy to train
- ✓ good at handling variable number of speakers
- ✓ scale (almost) linearly with file duration

End-to-End Neural Diarization

- ✓ one single step
- ✓ overlapping speech as first class citizen
- ✗ difficult to train
- ✗ not so good at generalizing to previously unseen number of speakers
- ✗ does not scale with file duration

EEND-VC

the best of both worlds



Speaker diarization

a love loss story

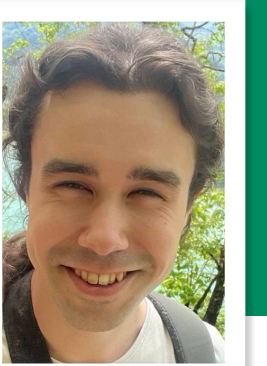
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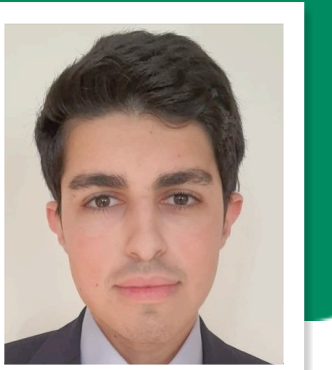
***Powerset multi-class cross entropy
loss for neural speaker diarization***

*Alexis Plaquet & Hervé Bredin
Interspeech 2023*



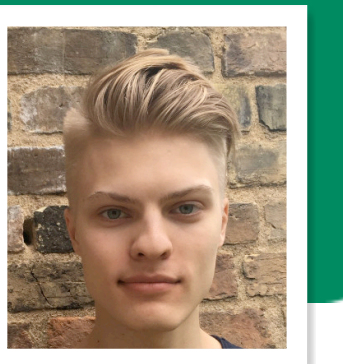
***Multi-latency look-ahead
for streaming speaker diarization***

*Bilal Rahou & Hervé Bredin
Interspeech 2024*



***PixIT: Joint Training of Speaker Diarization
and Speech Separation from Real-world
Multi-speaker Recordings***

*Joonas Kalda, Clément Pagés, Ricard
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Speaker Odyssey 2024*



Speaker diarization

a love loss story

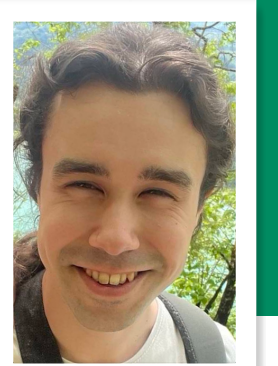
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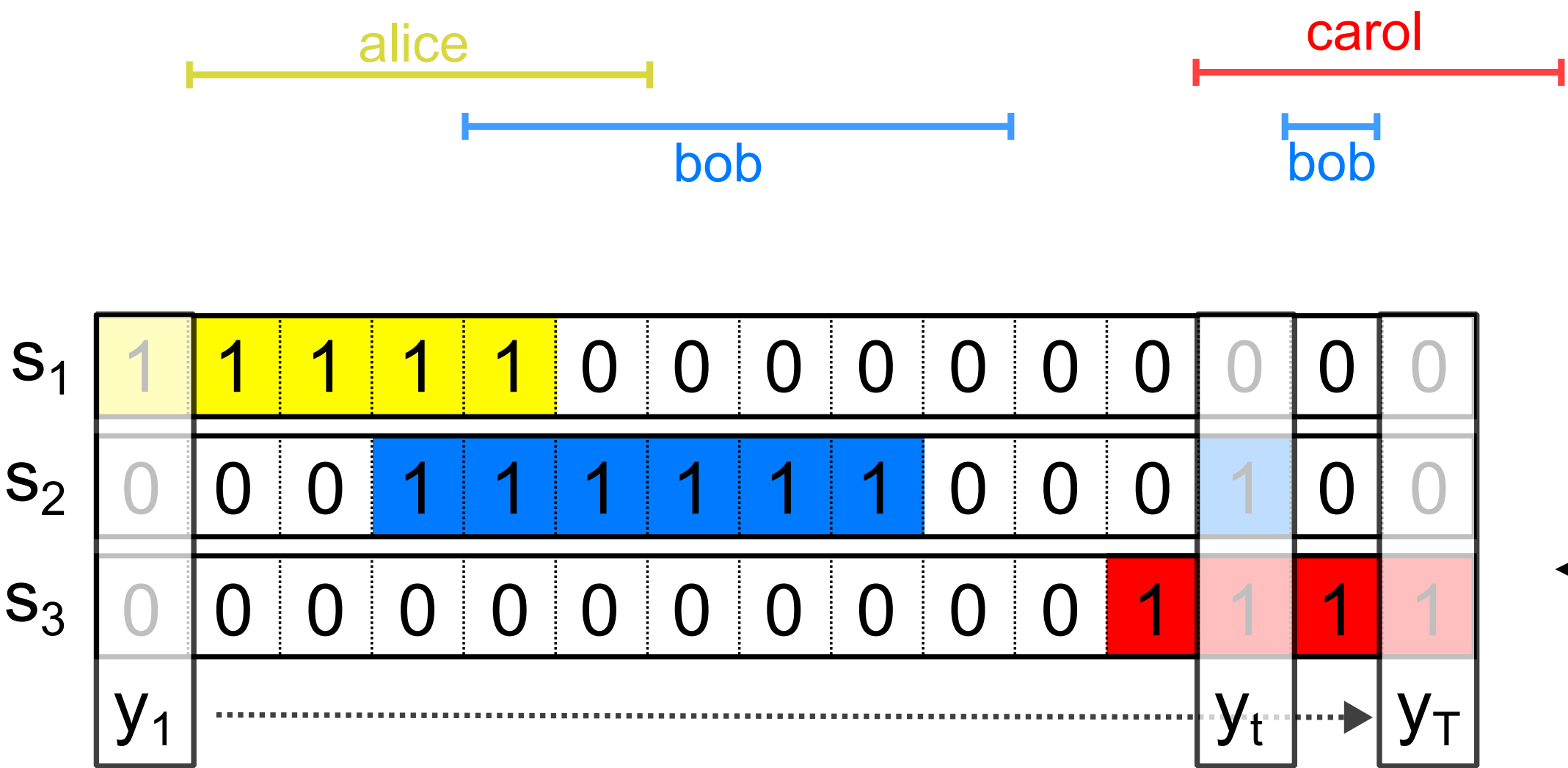
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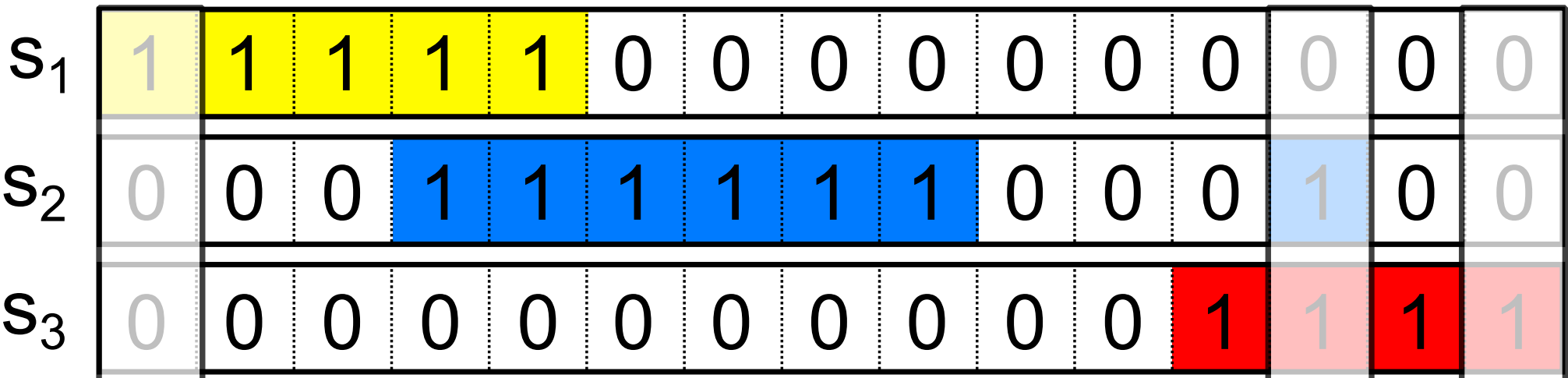
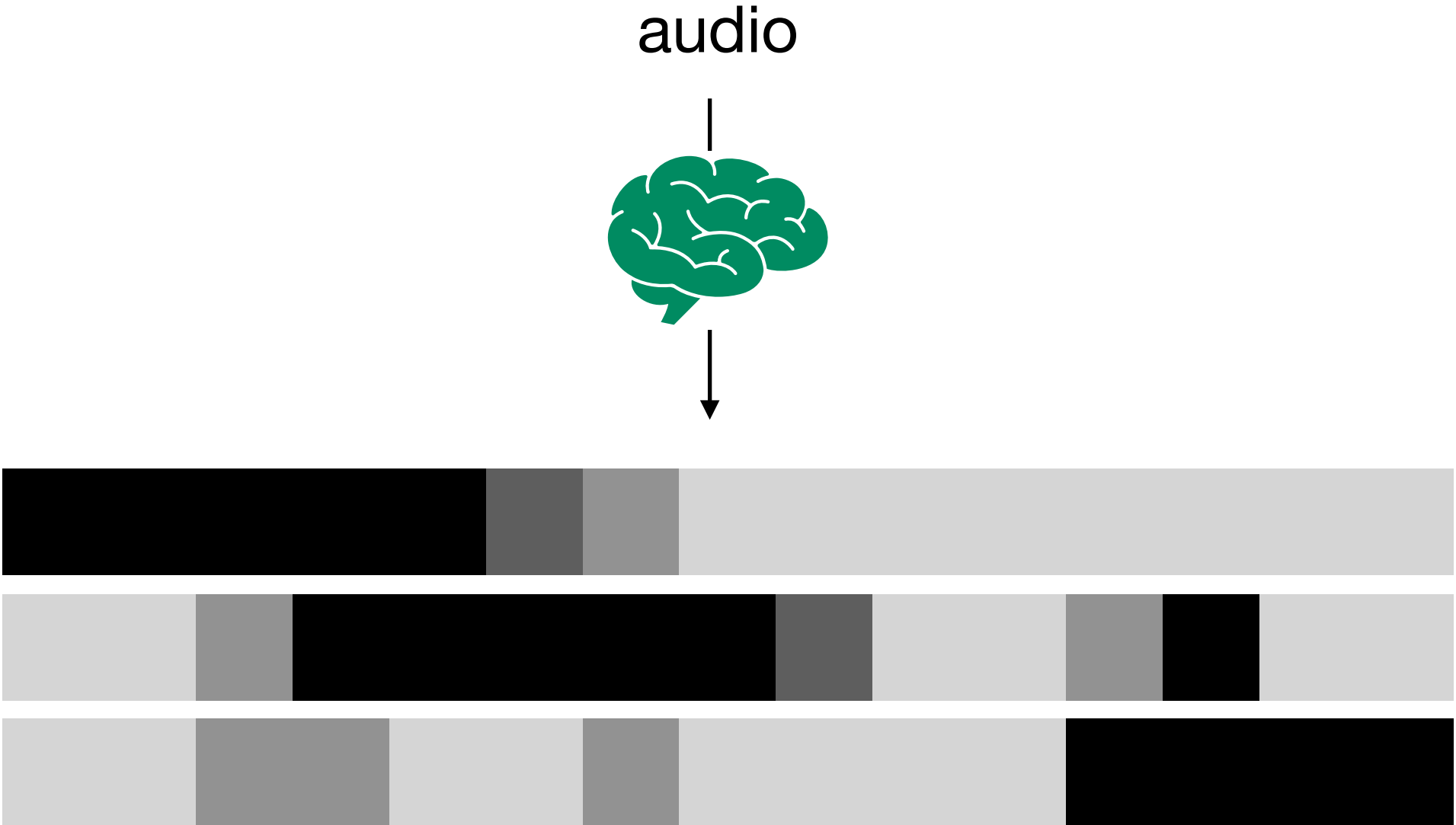
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End-to-End Neural Diarization

Frame-wise multi-label classification

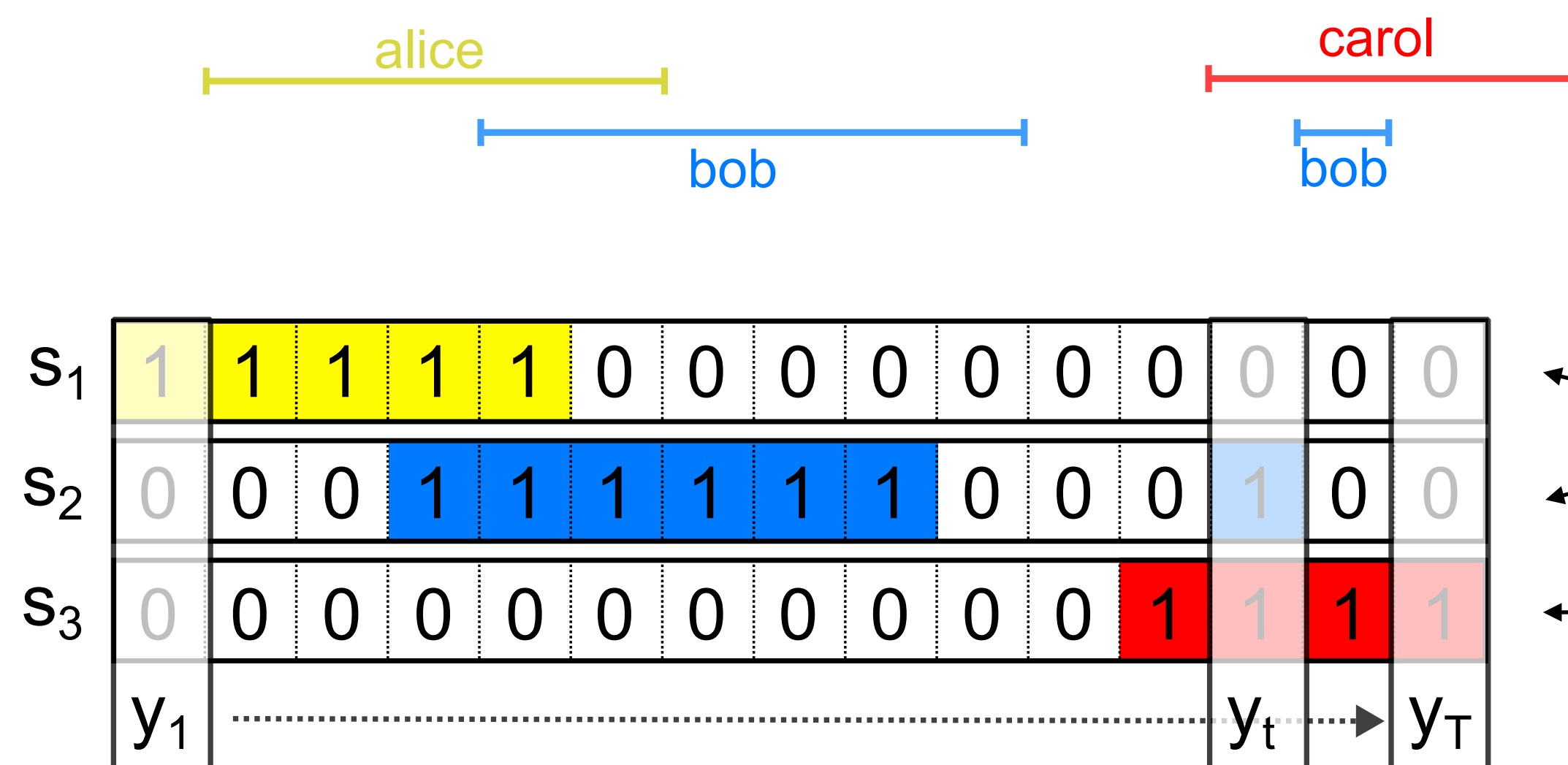


$\mathcal{L}_{BCE}$

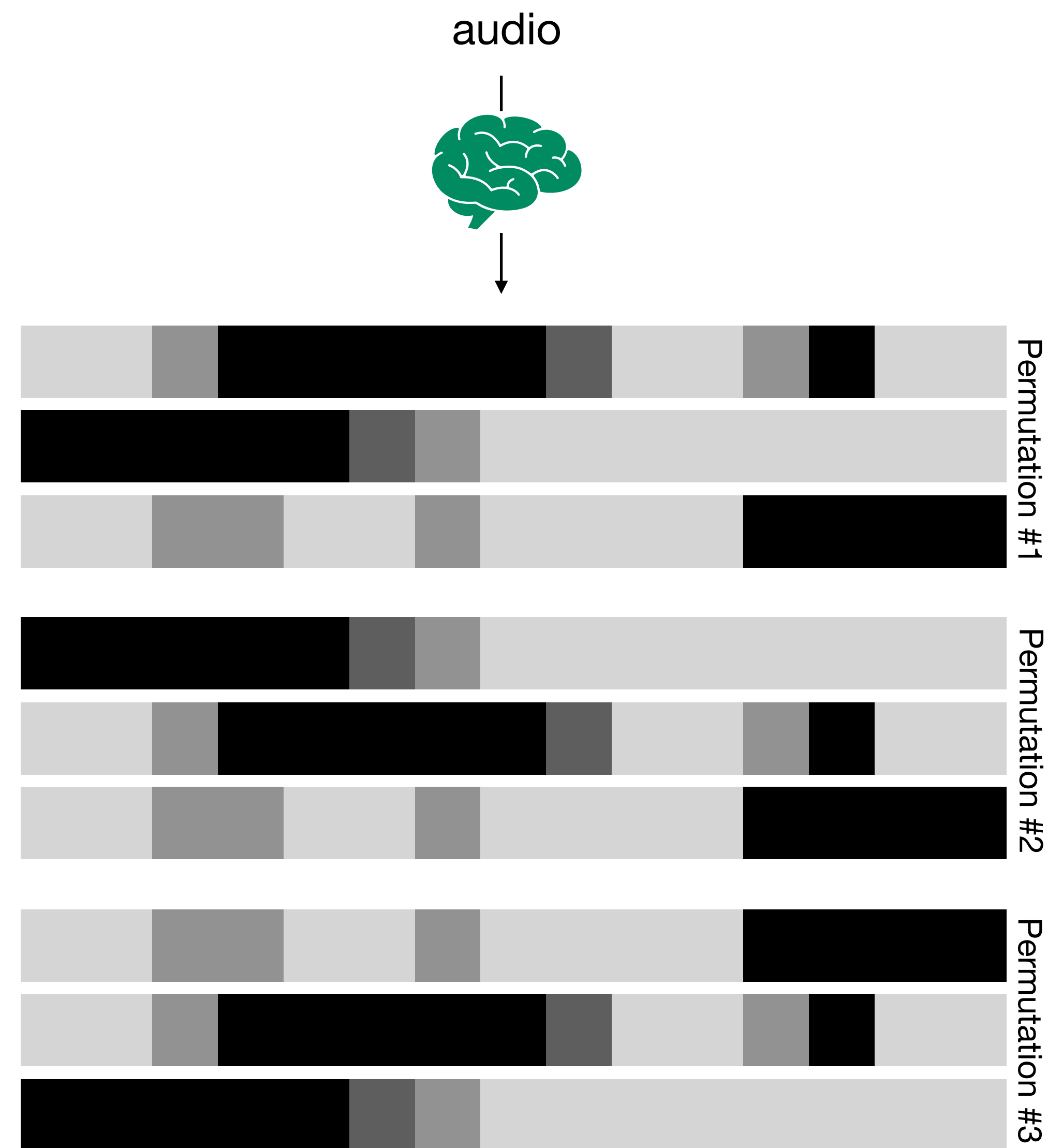


End-to-End Neural Diarization

Invariance to permutation



$$\mathcal{L} = \min_{p \in \mathcal{P}} \mathcal{L}_{BCE} (y, p(\hat{y}))$$

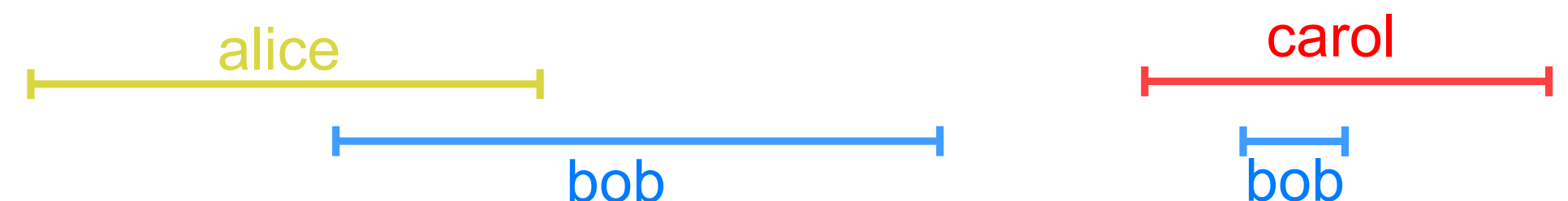


From multi-label to powerset encoding

- **dedicated multi-speakers classes**
number of classes growing fast
- **"regular" multi-class classification**
mutually exclusive classes
cross-entropy loss (easier to train)
- **thresholding replaced by argmax**
one less hyper-parameter to tune
- **dedicated non-speech class**
to control VAD aggressiveness

Multi-label encoding

s_1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0
s_2	0	0	0	1	1	1	1	1	1	0	0	0	1	0	0
s_3	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
y_1													y_t	$\rightarrow y_T$	

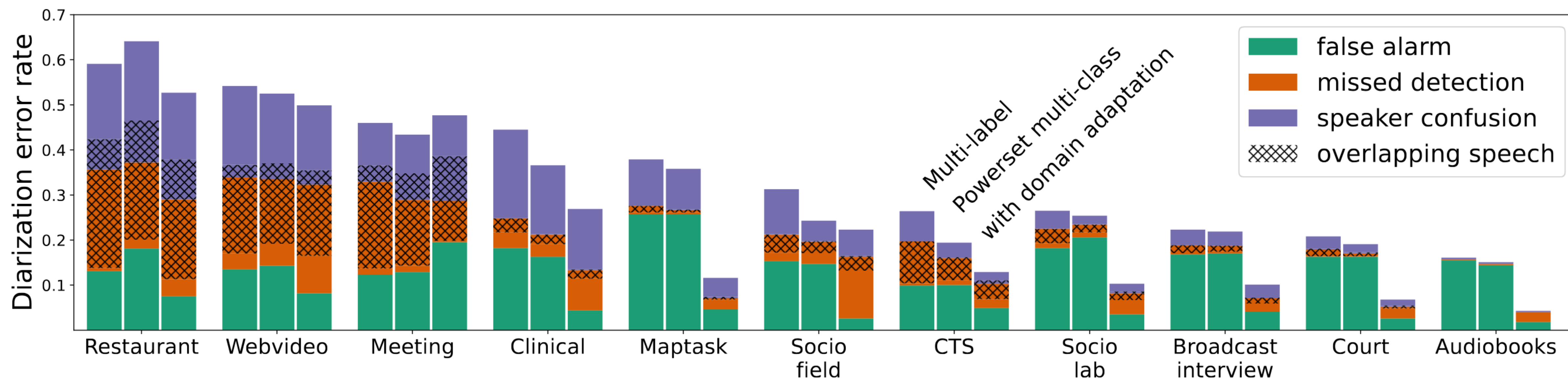


	y_1													y_t		y_T
$\emptyset$	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
$\{s_1\}$	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
$\{s_2\}$	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0
$\{s_3\}$	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1
$\{s_1, s_2\}$	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0
$\{s_1, s_3\}$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\{s_2, s_3\}$	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

Powerset multi-class encoding

Results

DIHARD 3 benchmark (11 application domains)



$$\text{DER} = \frac{\text{false alarm} + \text{missed detection} + \text{speaker confusion}}{\text{total speech}}$$

Speaker diarization

a love loss story

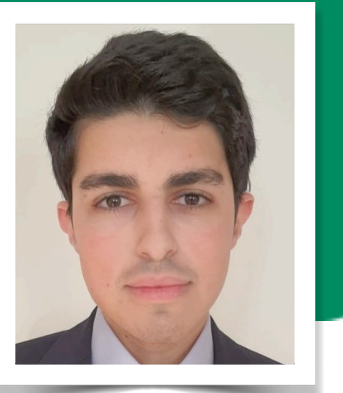
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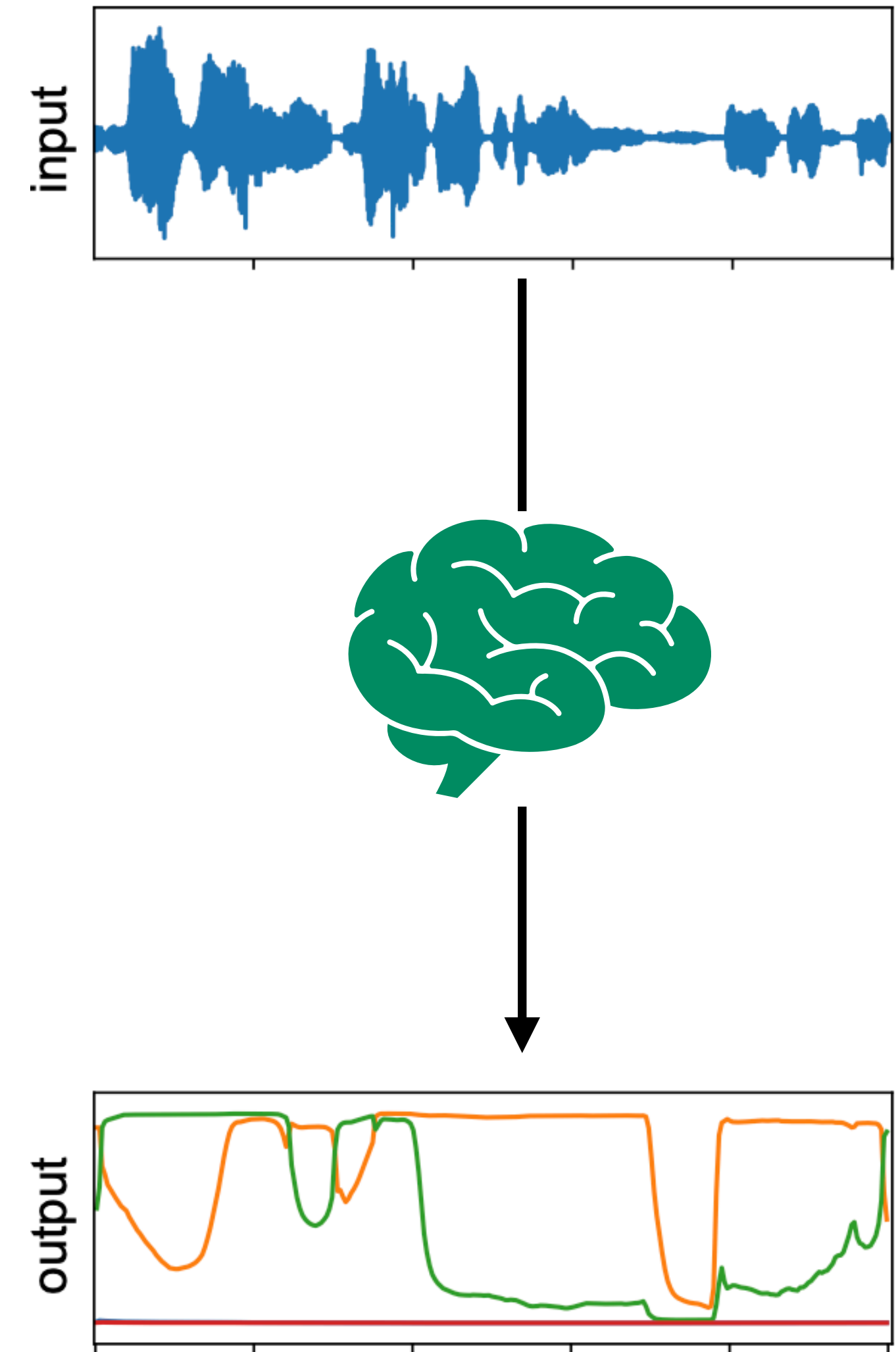
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Speaker Odyssey 2024*

Streaming speaker segmentation

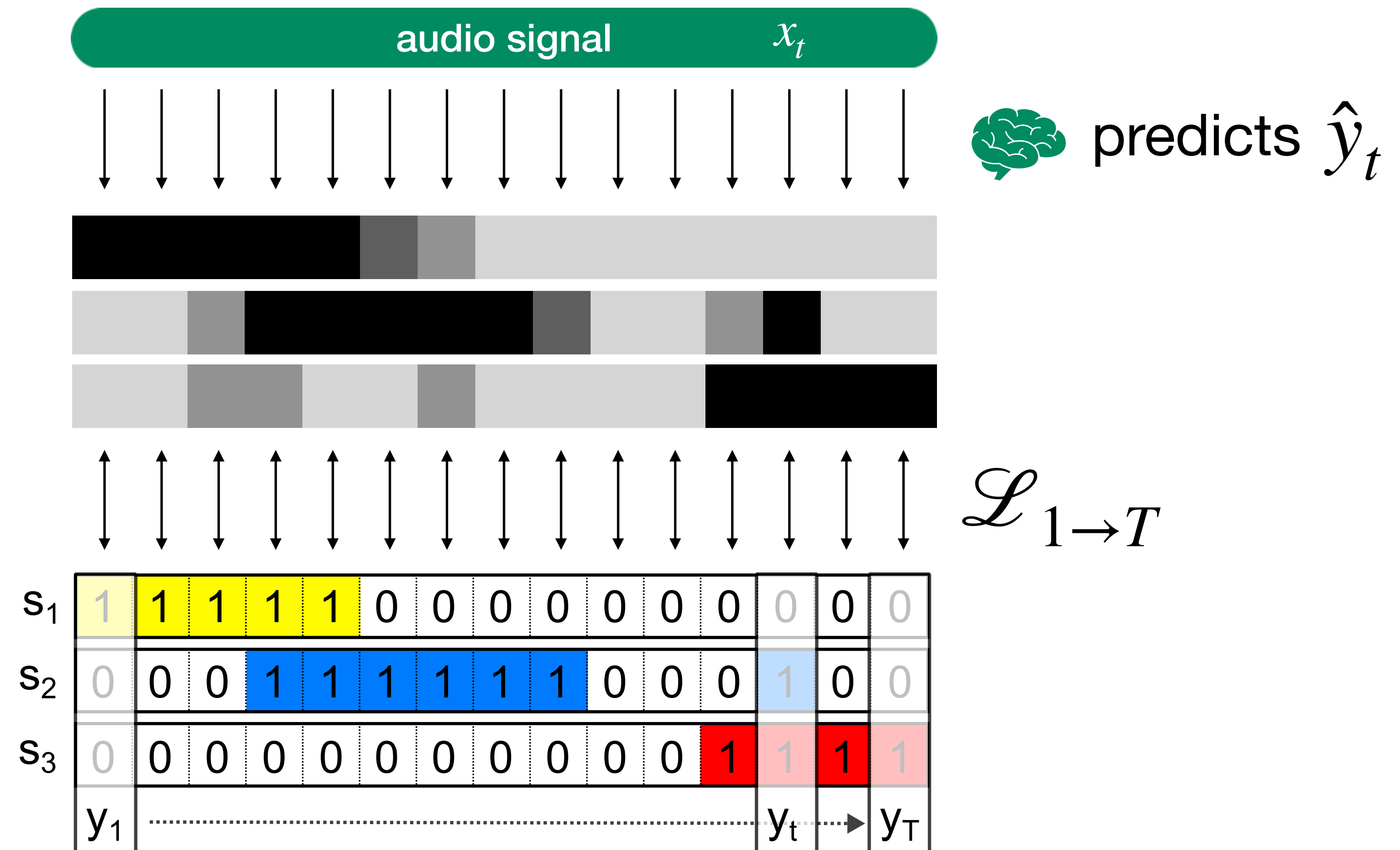
Causal architecture?

- Offline DER = 17.3%
- Sequence modeling
 - bi-directional LSTM ✗ 20.3%
 - mono-directional
- Feature extraction
 - Sincnet trainable filterbank (with instance normalization) ✗ 21.1%
 - no normalization
- Final classification
 - frame by frame ✓ how to go back to 17.3% ?



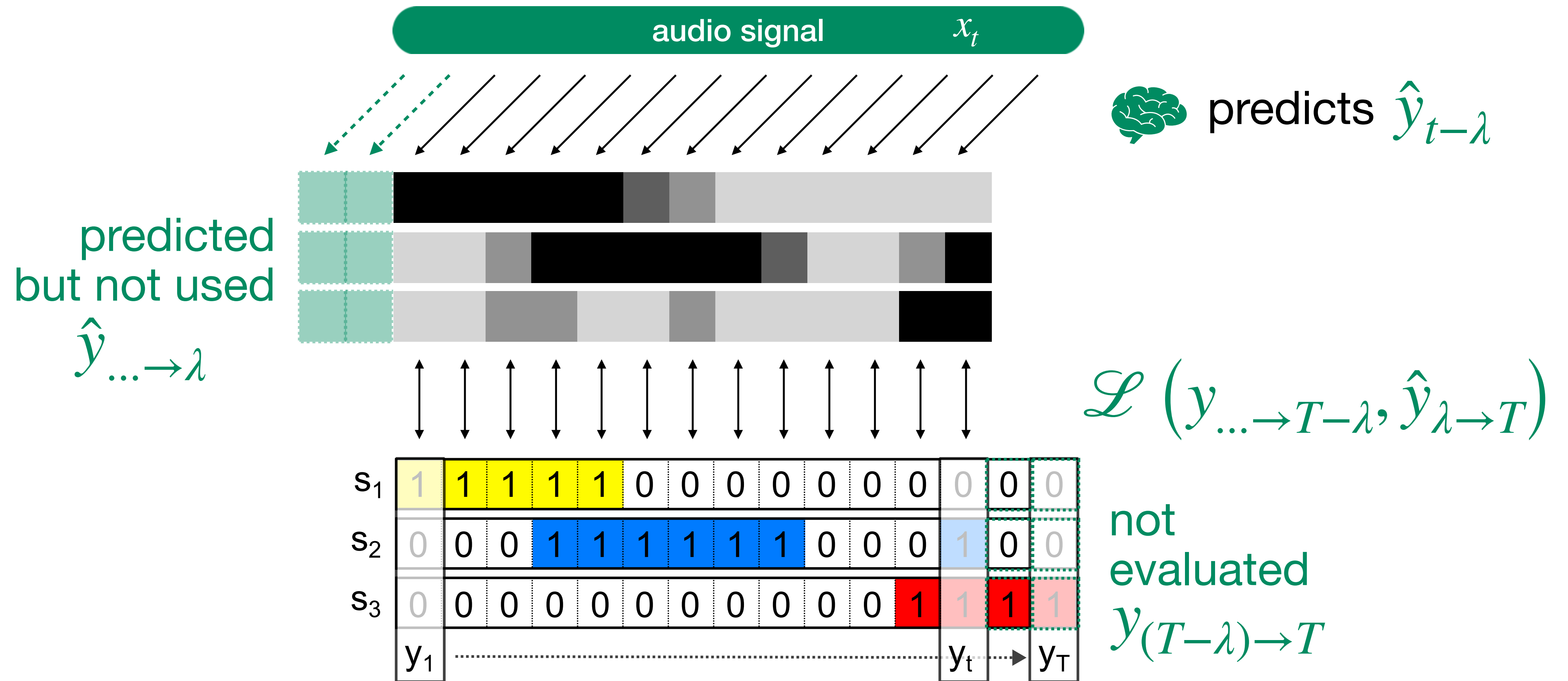
Streaming speaker segmentation

Look-ahead loss



Streaming speaker segmentation

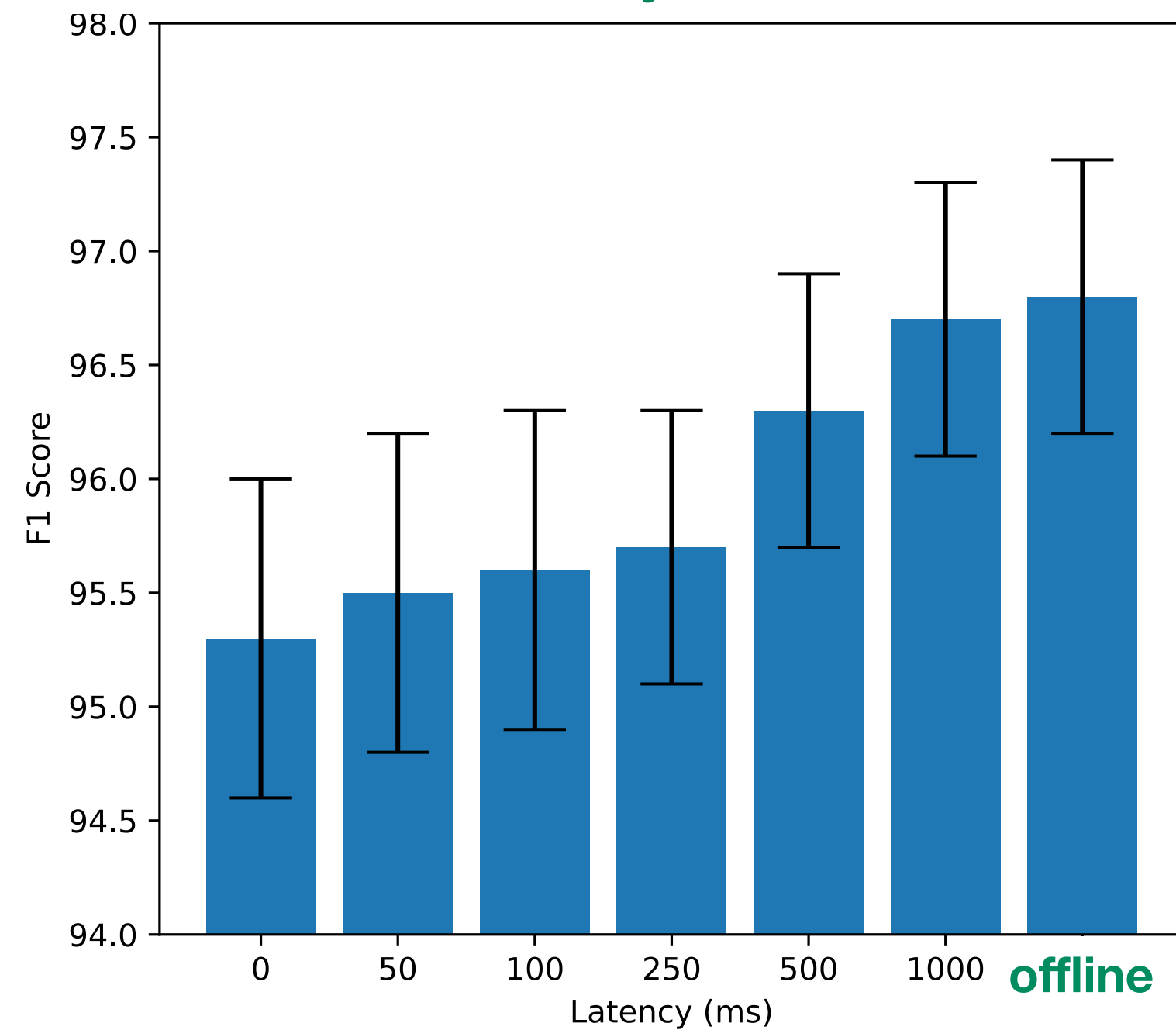
Look-ahead loss



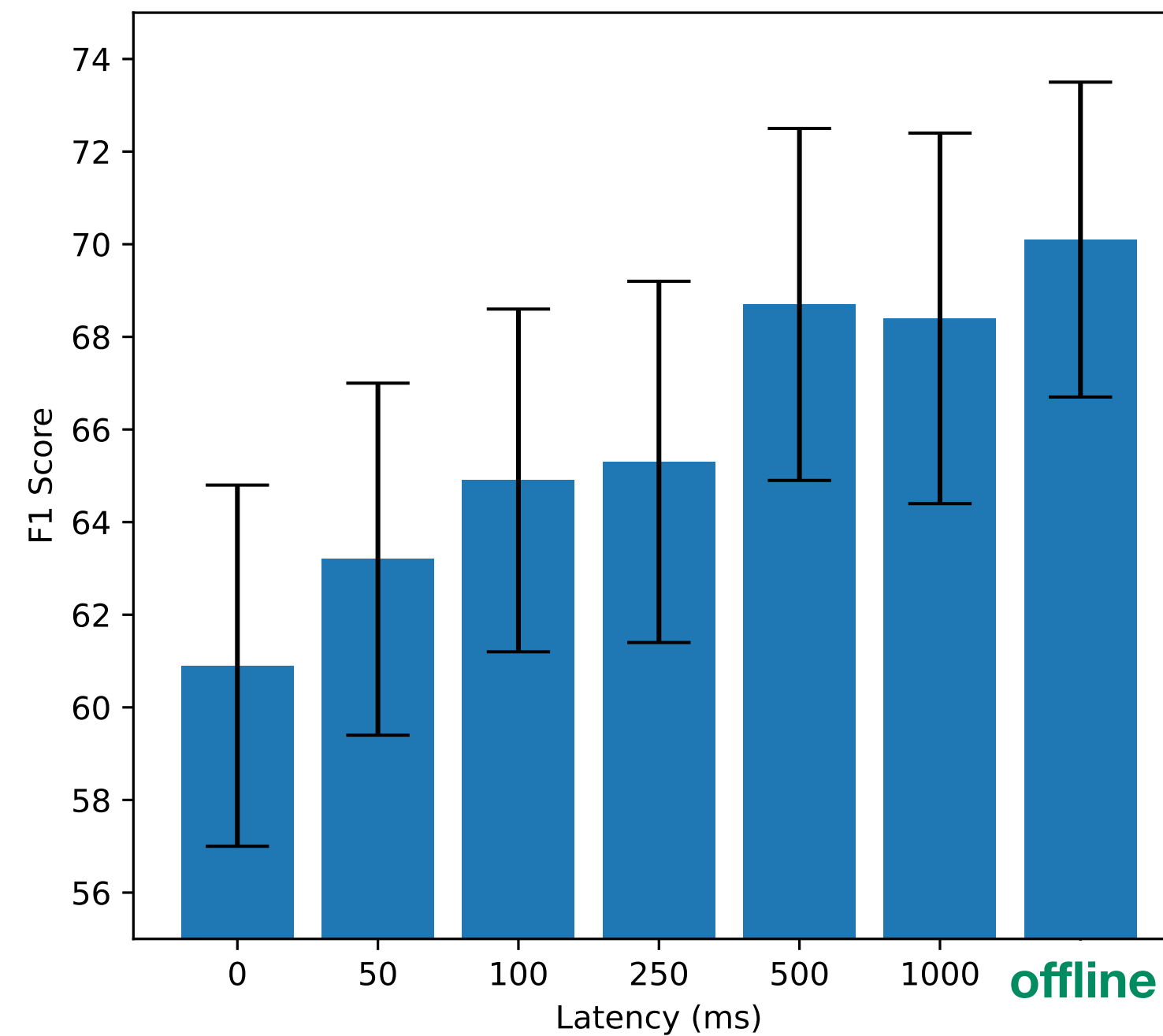
Streaming speaker segmentation

Latency/accuracy tradeoff

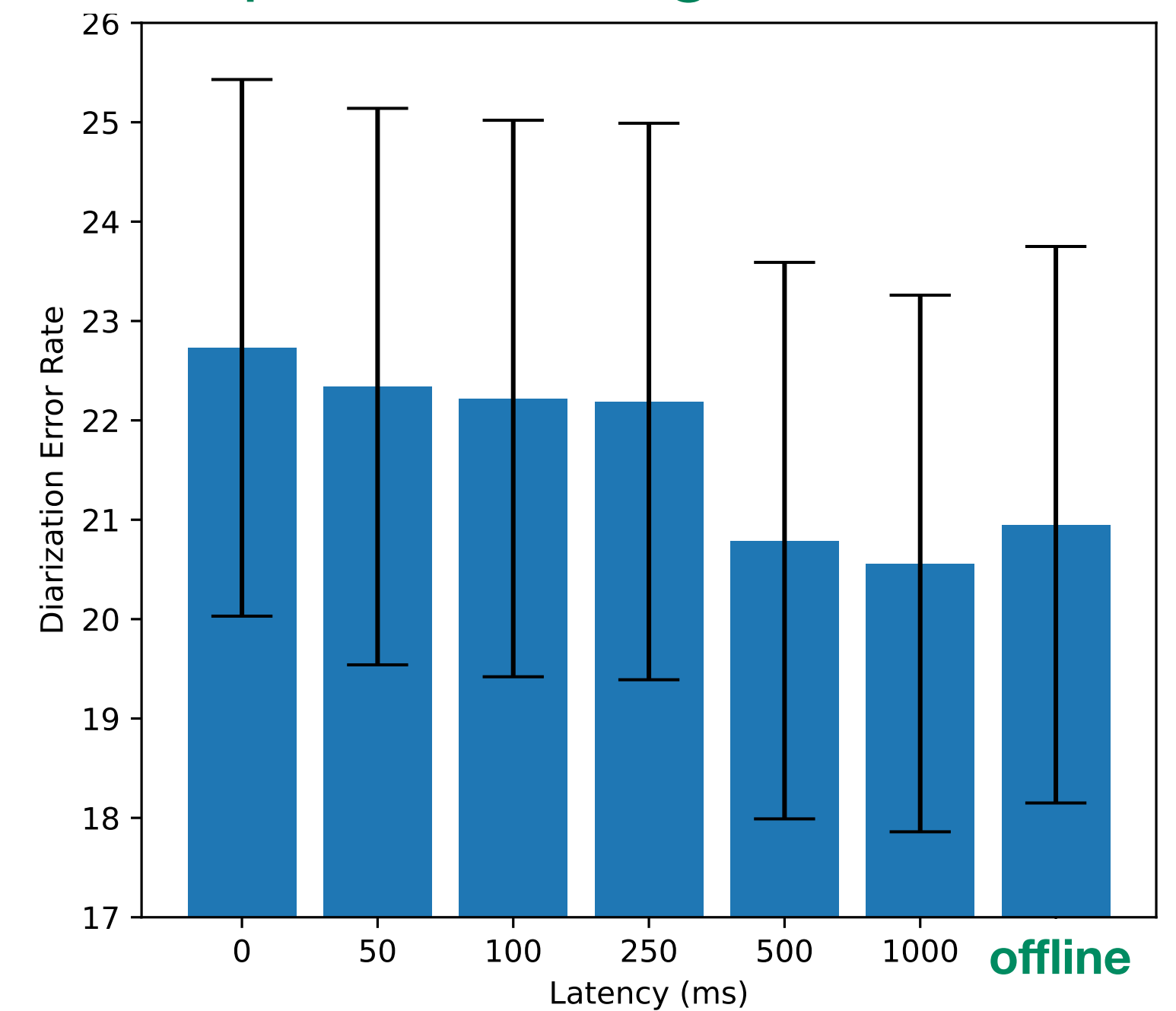
voice activity detection



overlapping speech detection

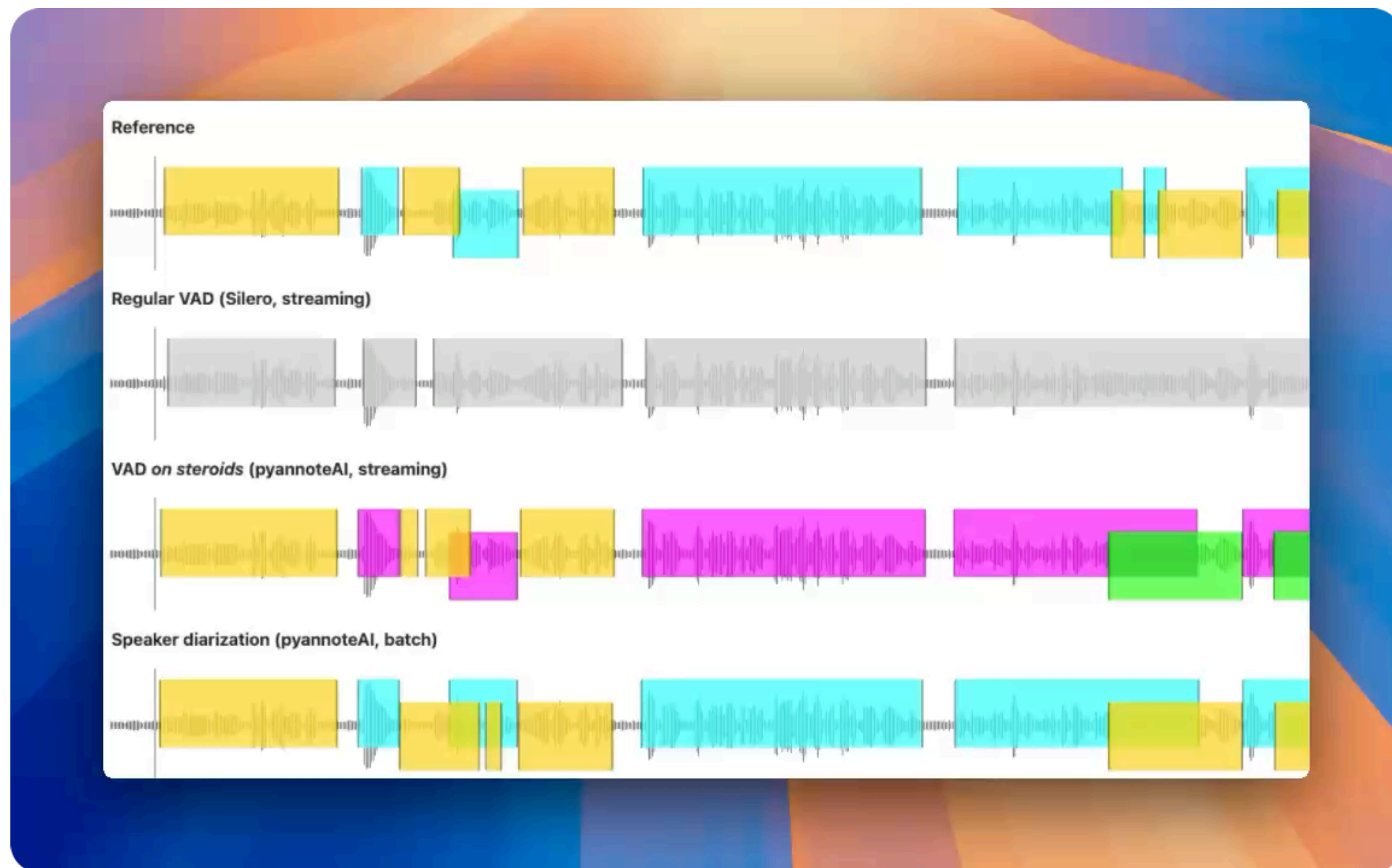


speaker change detection



Streaming speaker segmentation

VAD on steroids



Speaker diarization

a love loss story

Hervé Bredin



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some figures
courtesy of
Joonas Kalda

*Powerset multi-class cross entropy
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Interspeech 2023*

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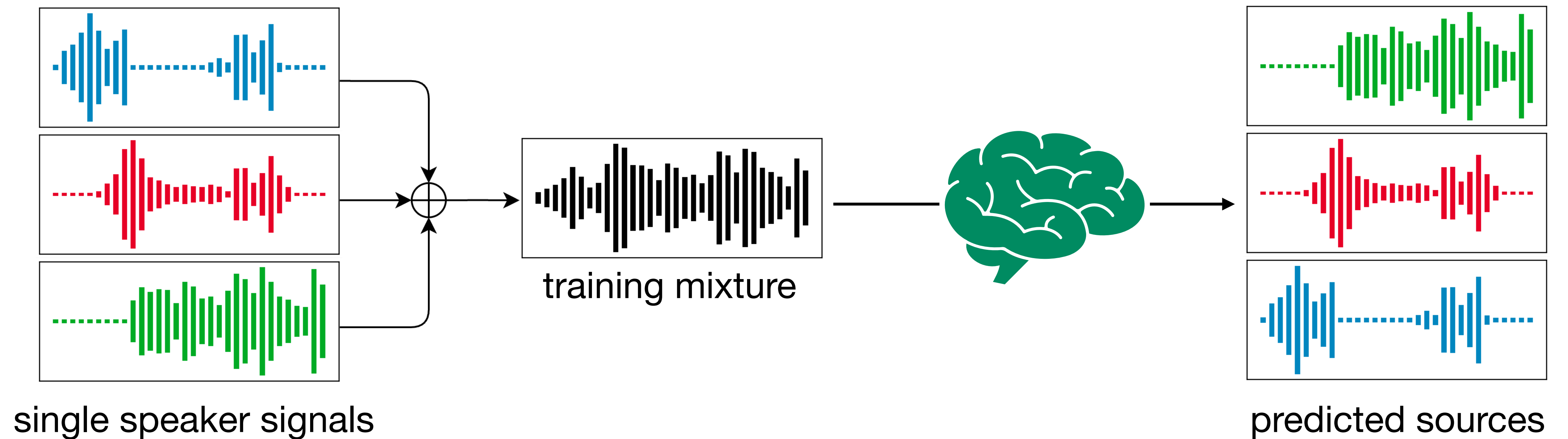
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Speech separation

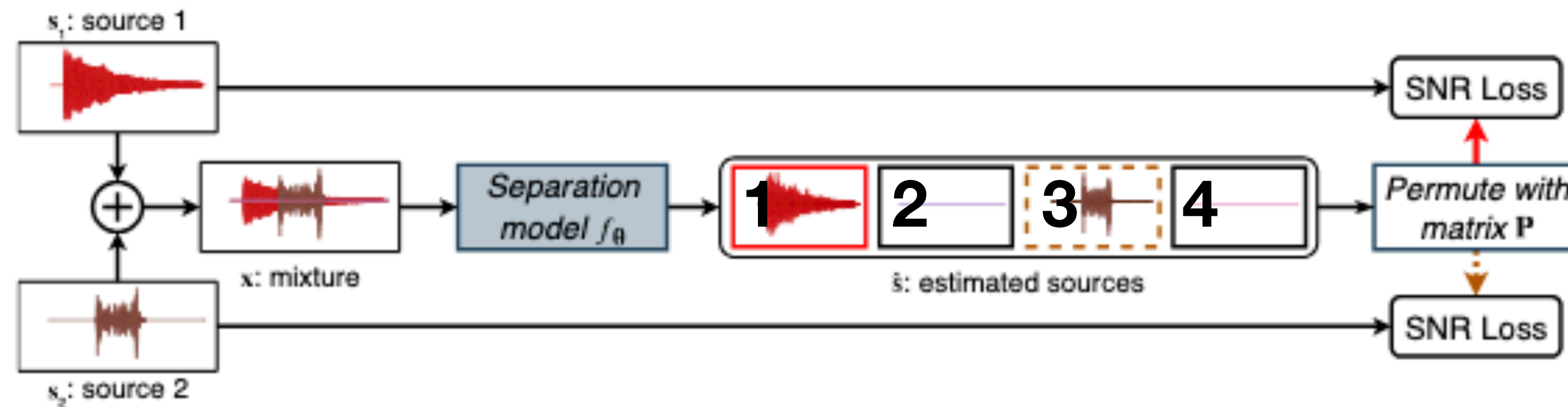
Supervised permutation-invariant training (PIT)



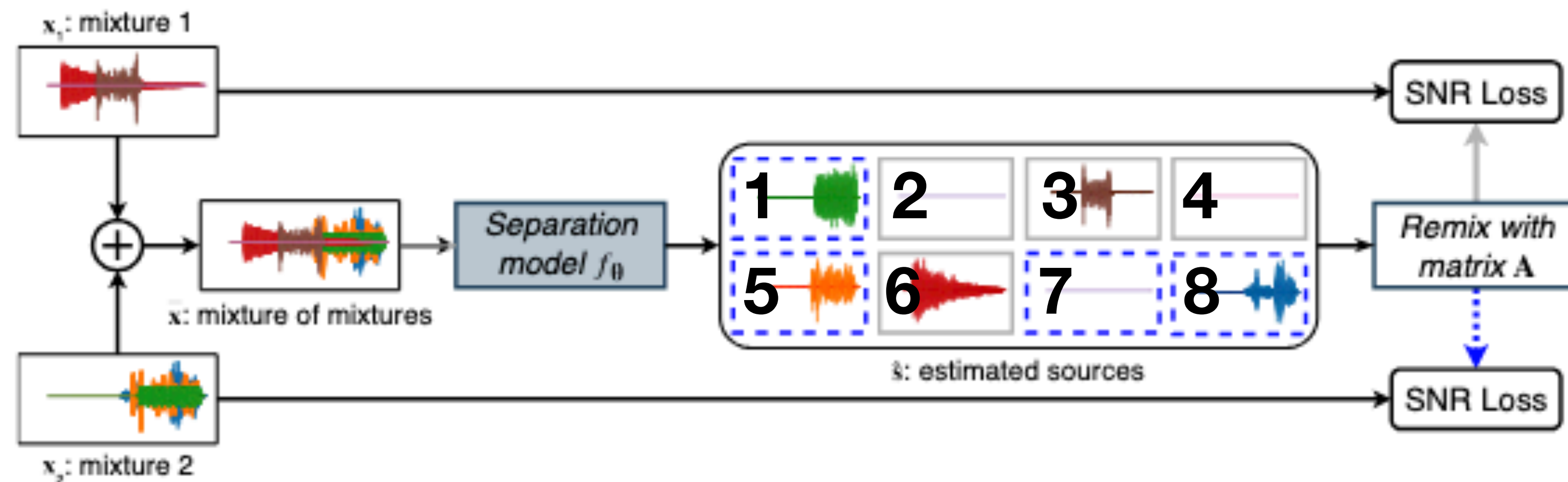
Speech separation

Unsupervised mixture-invariant training (MixIT)

"Unsupervised Sound Separation using Mixture Invariant Training"
Wisdom et al. NeurIPS 2020



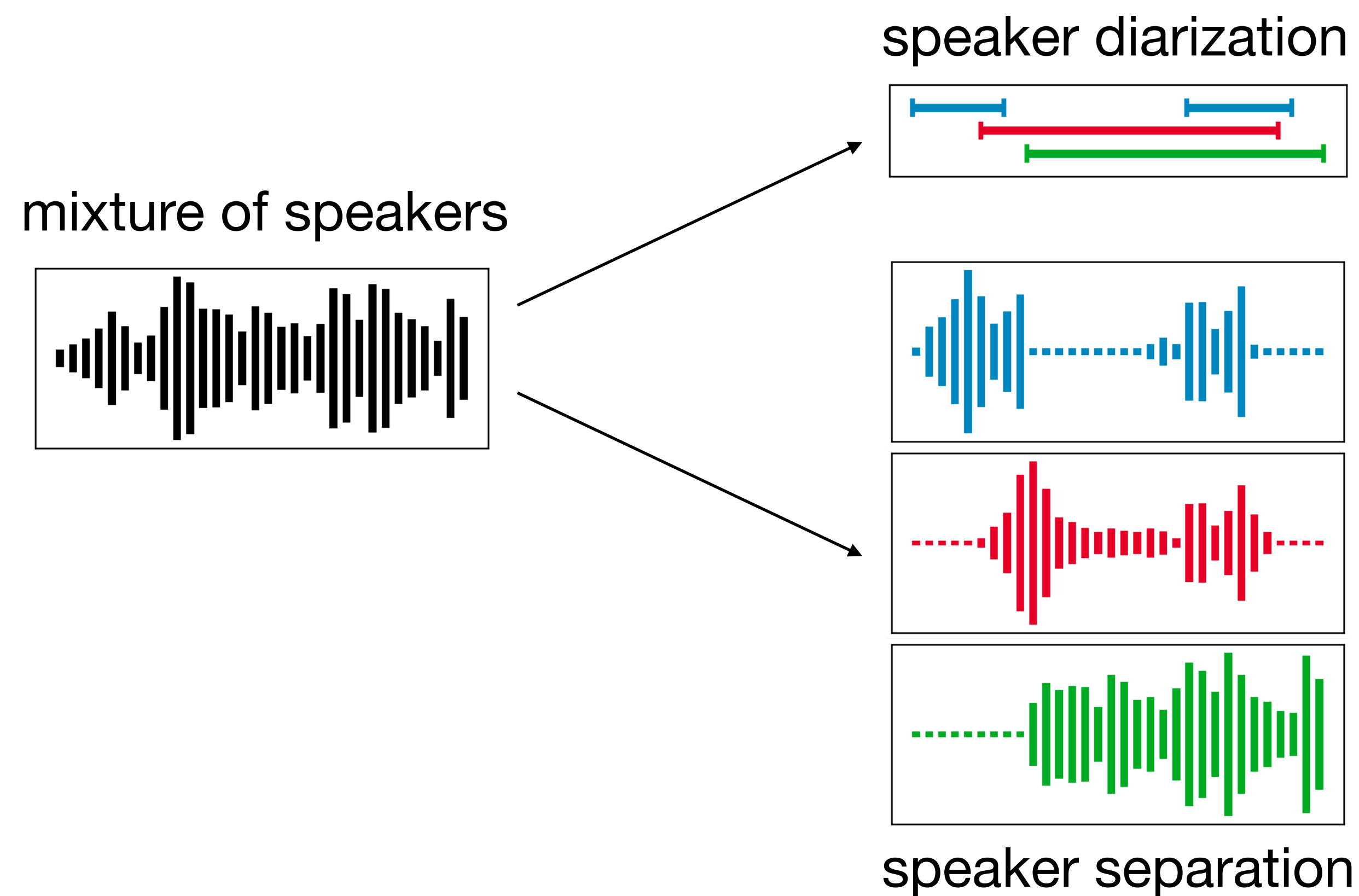
(a) Supervised permutation invariant training (PIT).



(b) Unsupervised mixture invariant training (MixIT).

Joint speaker diarization & separation

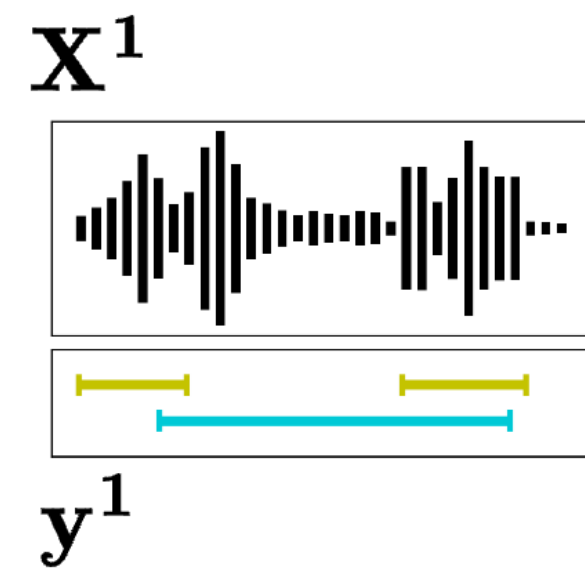
two complementary tasks



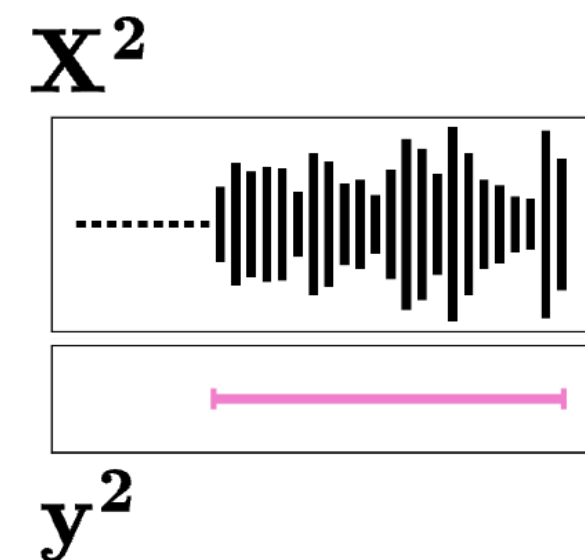
- **speaker activations** can be used to guide separation
- **separated sources** can help assign overlapping speech to the right speakers
- why not **train them jointly** in a **semi-supervised** manner?
 - supervised diarization
 - unsupervised separation

Joint speaker diarization & separation

PIT + MixIT = PixIT loss

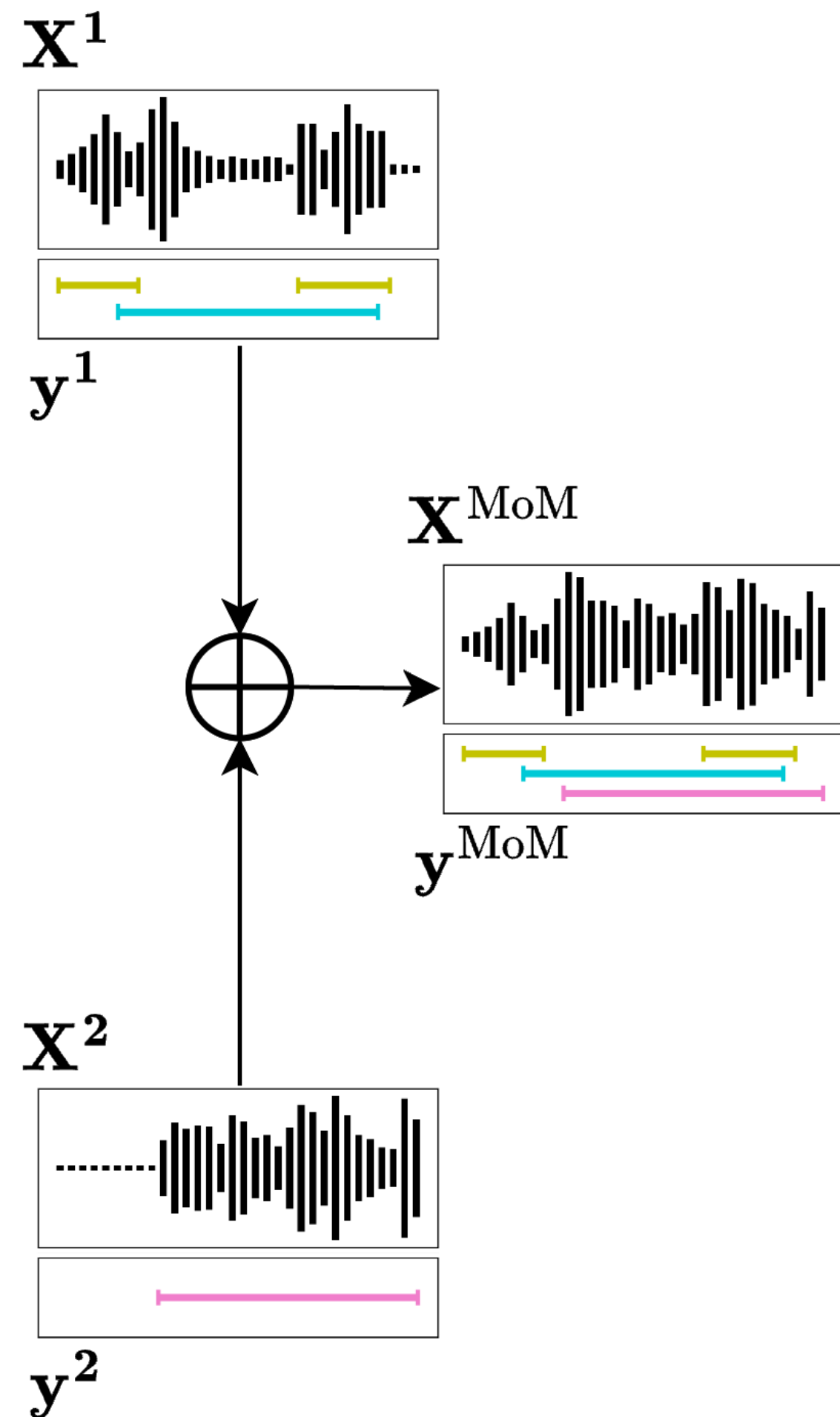


5 seconds chunks
with diarization labels



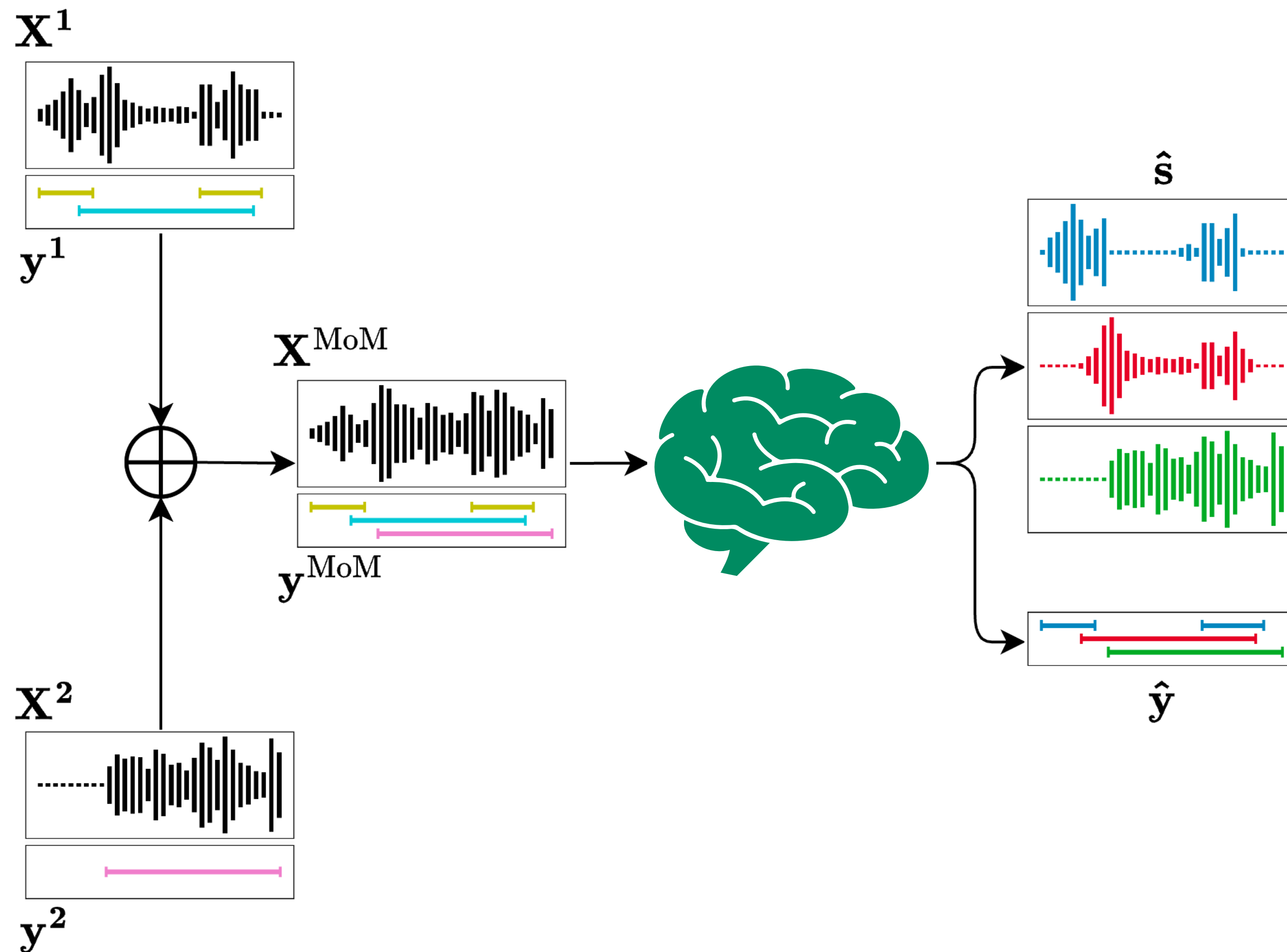
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PIT + MixIT = PixIT loss



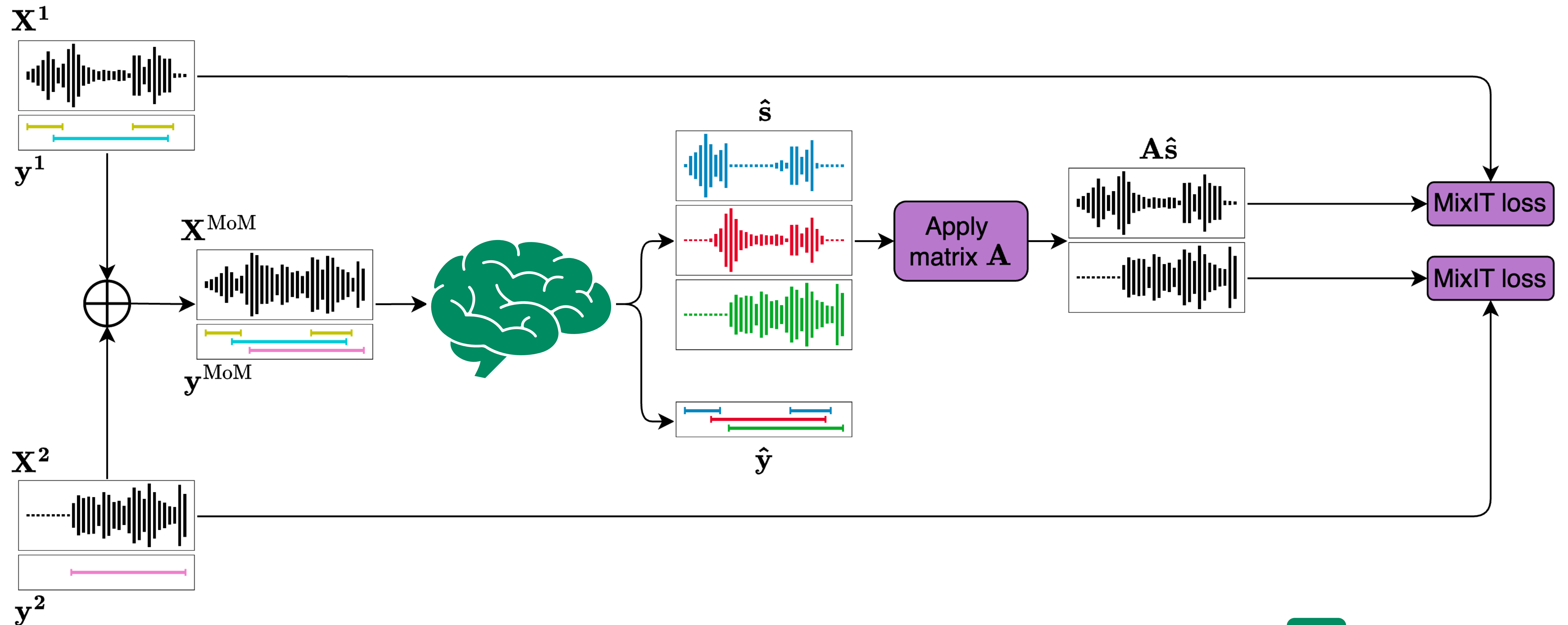
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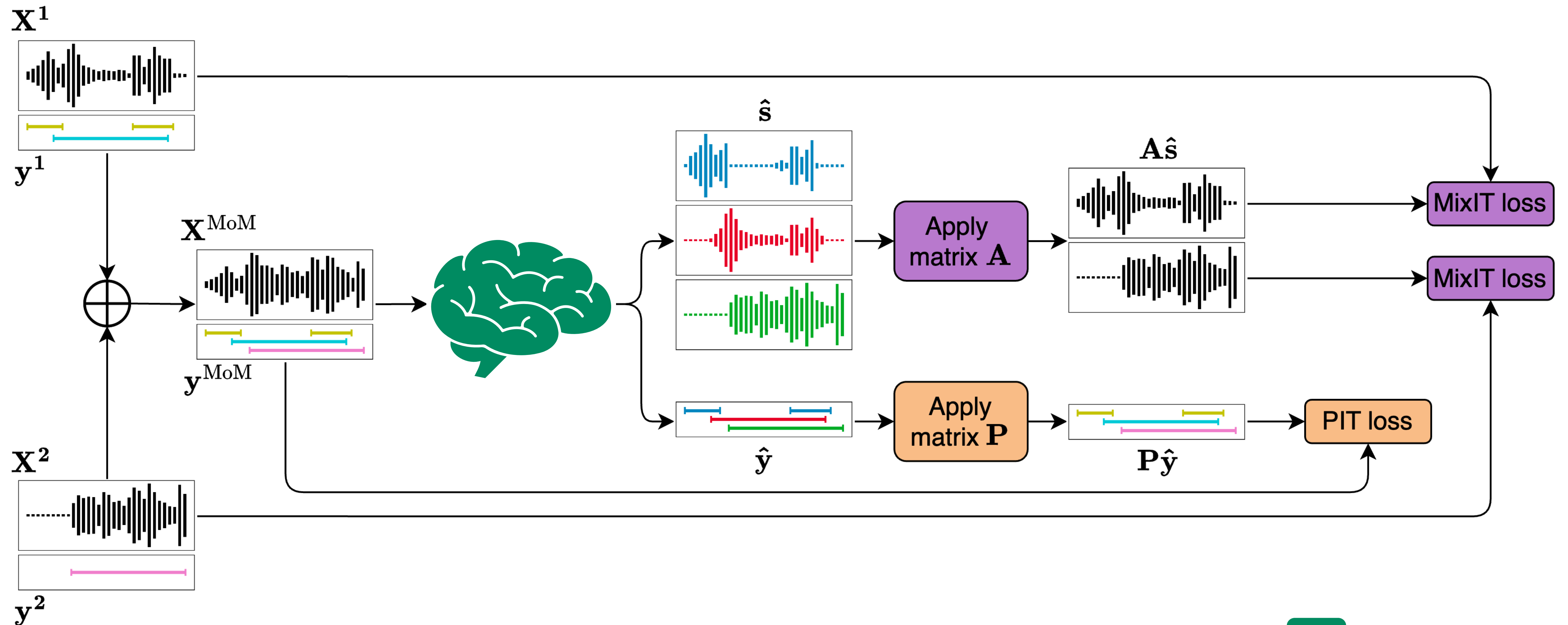
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Joint speaker diarization & separation

PIT + MixIT = PixIT loss

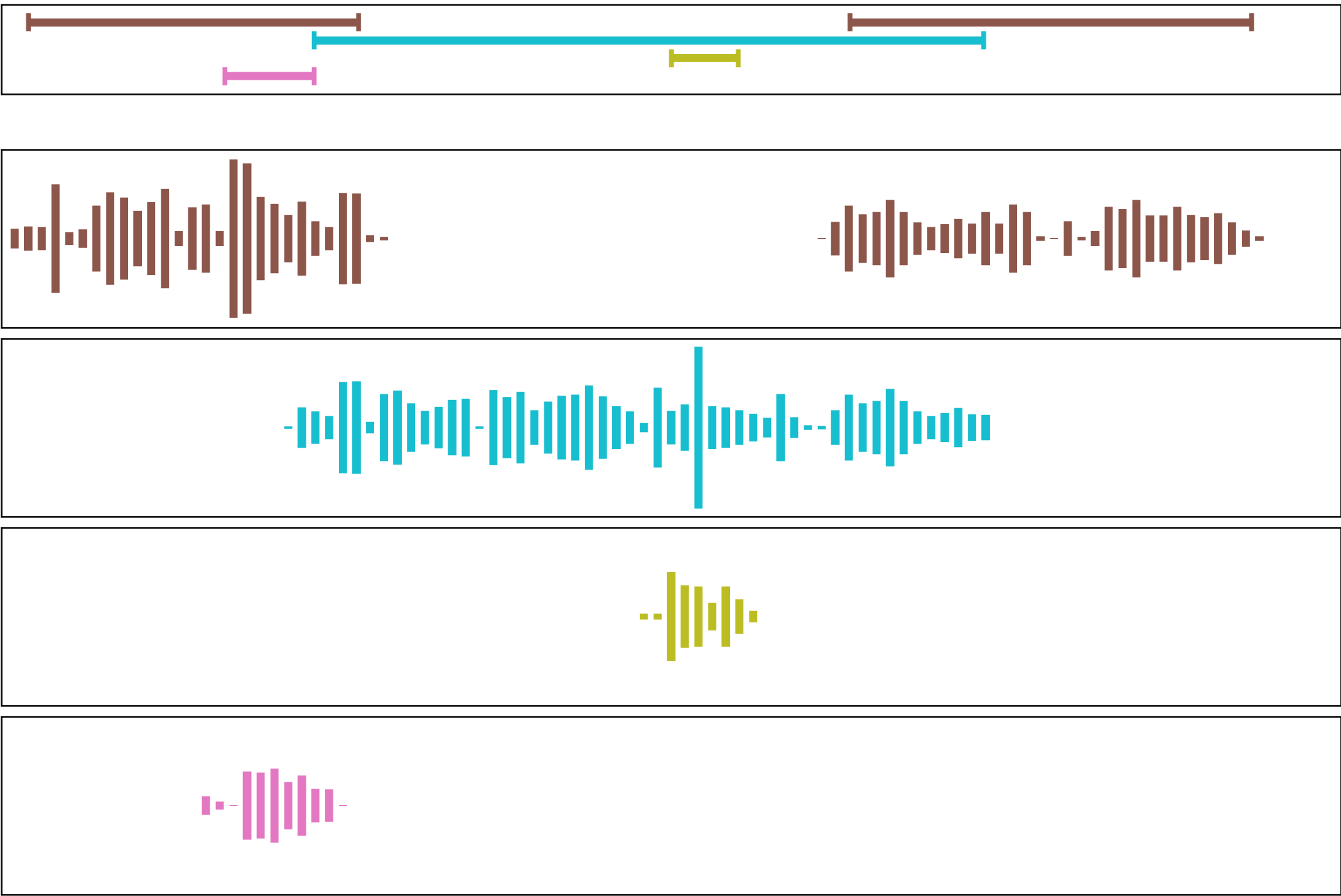


Inference on long-form audio



Diarization & separation

Inference on long-form audio



stitched, and transcribed separately

accounts for transcription
and diarization errors

Table 2: The cpWER (%) on AMI-SDM for various ASR systems with speaker attribution (SA) done through diarization or the joint model

ASR model	SA method	SA system	cpWER(%)				Relative Change
			sub	del	ins	total	
Whisper small.en	Diarization	pyannote 3.1	8.7	27.2	3.7	39.6	
	Diarization	PixIT	8.5	27.3	2.1	37.9	-4.3%
	Separation	PixIT	6.7	25.8	1.4	33.9	-14.4%
Whisper medium.en	Diarization	pyannote 3.1	7.4	28.0	3.4	38.8	
	Diarization	PixIT	7.3	27.8	2.0	37.1	-4.4%
	Separation	PixIT	5.9	25.8	1.2	32.8	-15.4%
Whisper large-v2	Diarization	pyannote 3.1	7.1	29.3	1.8	38.3	
	Diarization	PixIT	6.9	26.6	2.1	35.7	-6.7%
	Separation	PixIT	5.6	24.7	1.3	31.7	-17.2%
NeMo conformer large	Diarization	pyannote 3.1	11.5	36.0	1.4	48.9	
	Diarization	PixIT	13.3	33.9	1.3	48.5	-0.8%
	Separation	PixIT	13.4	24.6	1.4	39.4	-19.4%



pyannoteAI

Democratizing seamless human interaction,
grounded in scientific excellence



All things "speaker"

HIRING

- speaker diarization
- speaker identification
- speaker separation
- speaker-attributed transcription
- streaming speaker diarization
- streaming speaker identification
- interactive speaker diarization



Resources

 **brains**

Hervé Bredin, PhD
CSO, creator of pyannote.audio

Juan Coria, PhD
CTO, creator of diart

Antoine Laurent, PhD
researcher, co-inventor of
pyannote diarization model

your name goes here

 **compute**



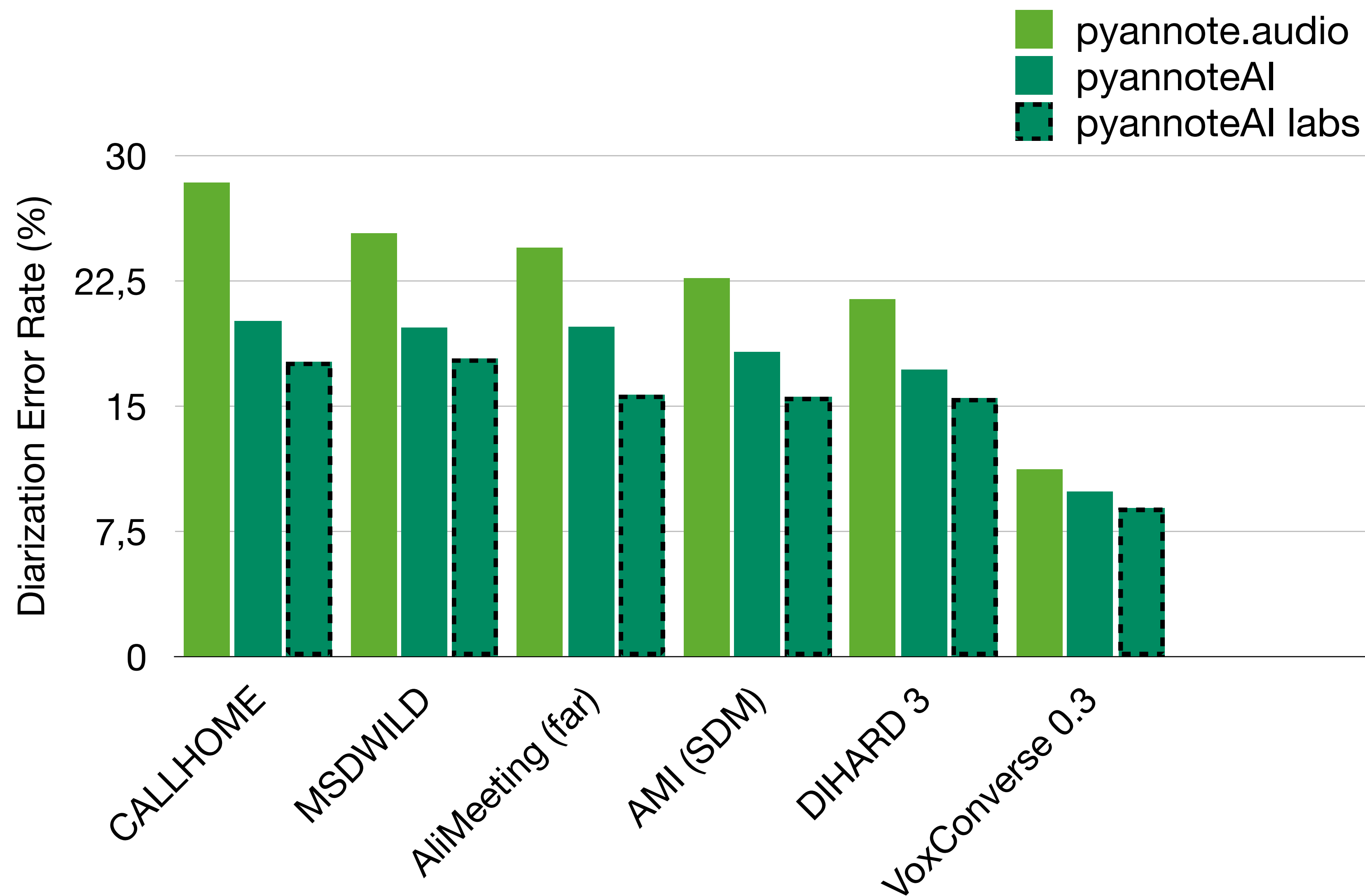
Jean Zay supercomputer
364 x 4 x H100 80Go

 **money**



Speaker diarization

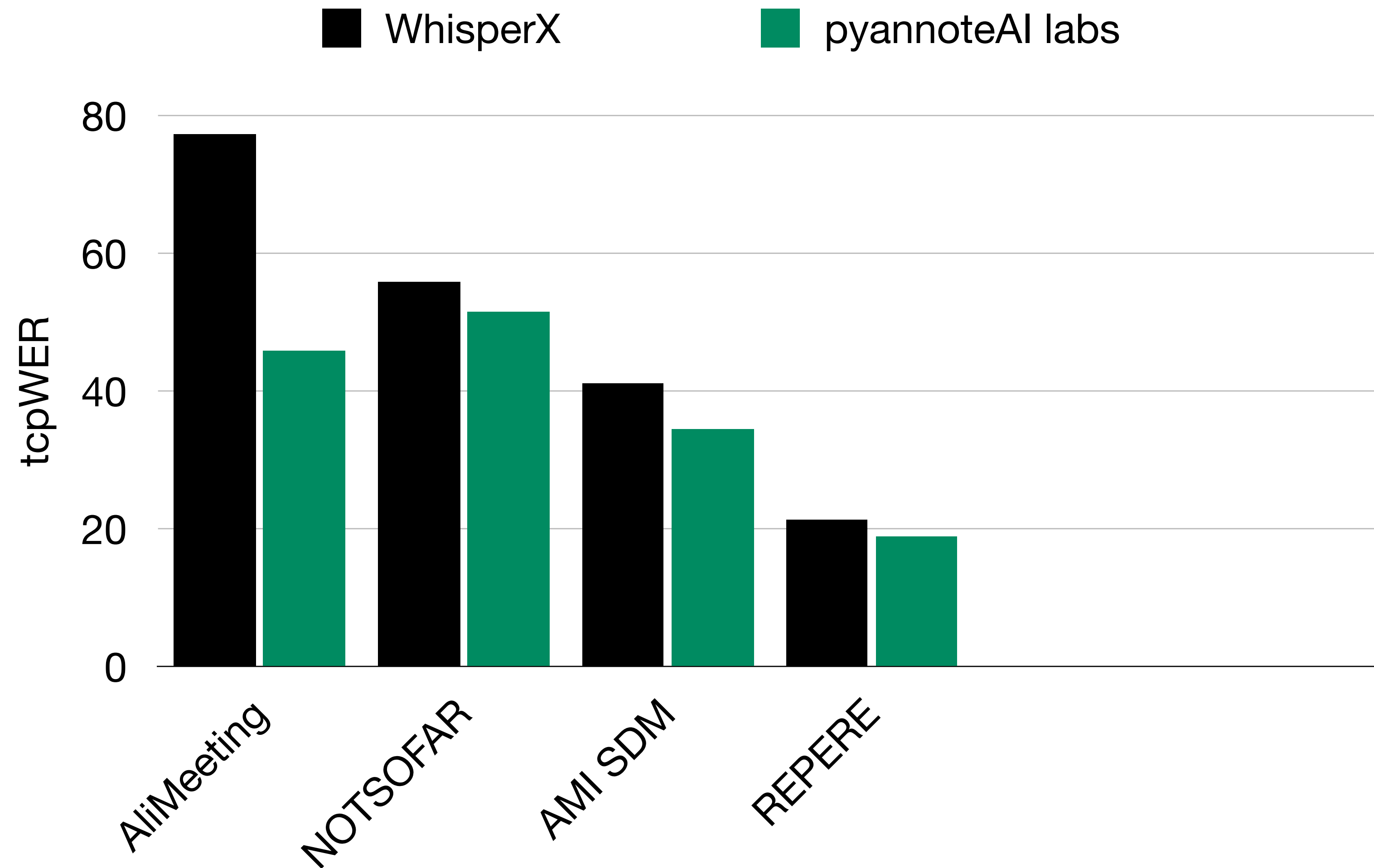
Without
finetuning





Speaker-attributed transcription

Off the shelf
faster-whisper-turbo





pyannoteAI

Hervé Bredin
herve@pyannote.ai